

The Path-Shadowing RMSE Discrepancy

Why the reproducibility report and the repository README disagree,
and what the number actually measures

results/Heston/preprocessing_with_log_returns
BASSERAS/benchmark

31 July 2026

Contents

1	The short answer	2
2	How the RMSE is computed	2
2.1	From ensemble to per-path error	2
2.2	The two aggregations	2
2.3	What the code does now	3
2.4	A naming defect — and what each row now reports	3
3	Why the difference exists	4
4	Three defects in the reproducibility report	4
5	A mixed-convention defect in the tables — found and fixed	5
6	The five cross-method tables	7
6.1	Bank size $N = 4\,096$	8
6.2	Bank size $N = 16\,384$	9
6.3	Bank size $N = 65\,536$	10
6.4	Bank size $N = 262\,144$	11
6.5	Bank size $N = 1\,000\,000$	12
7	Figures	13
7.1	Stylised-fact diagnostics (8-panel)	14
7.2	Path-shadowing convergence and calibration	22
7.3	Training and scoring diagnostics	29
7.4	Embedding visualisations (PCA / t-SNE)	32

1 The short answer

Nothing was recomputed and no bank changed. **The two documents apply a different final aggregation step to the same per-path errors**, and the CSDI horizon-realised-volatility (rv) row is where the gap is widest.

Source	Aggregation	CSDI rv RMSE @1M
Report Table 8 (§4.3, “originally published”)	$\sqrt{se}$ (root last)	0.01772
Report Table 5 (§3.5, “corrected”)	$\sqrt{se}$ (root inside)	0.01373
Repository README (current)	$\sqrt{se}$ (root inside)	0.01373

So the discrepancy you spotted is **internal to the reproducibility report**: it prints both conventions, in Table 8 and Table 5, without ever showing them side by side. The README agrees with Table 5 to every printed digit; it disagrees with Table 8 by a factor of 1.291. Only the RMSE rows differ — CRPS, coverage, width and miss-rate cells are identical between Table 8 and Table 5, because those metrics are plain arithmetic means and are untouched by the aggregation choice.

Read §2.4 before using the tables. The row quoted above is *rv*, and *rv* still reports root-inside — correctly, because for a scalar quantity that aggregation *is* a mean absolute error, and the tables now say MAE. The two *trajectory* quantities, cum and step, no longer do: as of 2026-07-31 they report the textbook root-last RMSE, so those rows in §6 match Table 8’s convention rather than Table 5’s. The scorer emits both aggregations for every model, bank and quantity, so nothing is lost and every “before” number in this document is read live from the artefacts.

2 How the RMSE is computed

2.1 From ensemble to per-path error

Fix a held-out query path $q \in \{1, \dots, m\}$, $m = 512$, and a horizon step $u \in \{1, \dots, H\}$. Path shadowing retrieves $K = 256$ nearest neighbours from the bank and forms an ensemble forecast $Y_{q,u,k}$. The squared error of the *ensemble mean* against the realised value $y_{q,u}$ is

$$se_{q,u} = \left(\frac{1}{K} \sum_{k=1}^K Y_{q,u,k} - y_{q,u} \right)^2,$$

and each path is reduced to a single scalar by averaging over the horizon,

$$se_q = \frac{1}{H} \sum_{u=1}^H se_{q,u}.$$

Everything above is common to both documents. **The disagreement is entirely in the last step**, which turns the length- m vector (se_q) into one reported number.

2.2 The two aggregations

$$\text{(A) root-last: } R_A = \sqrt{\frac{1}{m} \sum_{q=1}^m se_q} \qquad \text{(B) root-inside: } R_B = \frac{1}{m} \sum_{q=1}^m \sqrt{se_q}$$

(A) is the textbook root-mean-square error: one square root, taken last. (B) is the *mean per-path root error* — a square root per query path, then an ordinary average.

Because $x \mapsto \sqrt{x}$ is **concave**, Jensen’s inequality gives

$$R_B = \mathbb{E}[\sqrt{se}] \leq \sqrt{\mathbb{E}[se]} = R_A,$$

with equality if and only if every se_q is identical. **(B) is therefore always the smaller number**, and the gap widens as the spread of per-path errors grows. This is why the effect is largest on rv: a scalar quantity with no horizon averaging to smooth its error distribution.

2.3 What the code does now

The scorer (`path_shadowing_pdf.py`, `_agg`) computes *both* aggregations from the same per-path vector and writes both to every artefact:

```
if kind == "rmse":      return float(np.sqrt(per).mean())
if kind == "rmse_last": return float(np.sqrt(per.mean()))
```

`np.sqrt(per).mean()` roots *then* averages — convention (B), JSON key `rmse`.
`np.sqrt(per.mean())` averages *then* roots — convention (A), JSON key `rmse_textbook`.
The paired bootstrap carries the matching branch (`np.sqrt(samp).mean(axis=1)` versus `np.sqrt(samp.mean(axis=1))`), so each confidence interval is consistent with its own point estimate.

2.4 A naming defect — and what each row now reports

Under convention (B) the reported quantity **is not a root-mean-square error**, and the two kinds of quantity fail the name in different ways.

rv — reports (B), labelled MAE. The horizon collapses ($H = 1$), so $se_q = e_q^2$ and

$$R_B = \frac{1}{m} \sum_q \sqrt{e_q^2} = \frac{1}{m} \sum_q |e_q| = \textbf{MAE exactly}.$$

This row is a mean absolute error. The reproducibility report labels it “RMSE”; **the tables in §6 label it MAE**, which is simply its correct name. The value is unchanged — only the label is.

cum and step — report (A), labelled RMSE. Here $H = 32$ and (B) gives $\frac{1}{m} \sum_q \sqrt{\text{mean}_u se_{q,u}}$: an average of per-path horizon-RMS values. That is a hybrid with no standard name, and it is not an RMSE either. Calling it one understates the textbook figure by roughly 18% (cum) and 6% (step). Since se_q is *already* the mean over the horizon,

$$R_A = \sqrt{\frac{1}{m} \sum_q se_q} = \sqrt{\frac{1}{mH} \sum_q \sum_u se_{q,u}}$$

— the root of the mean squared error over every (query, horizon-step) pair, i.e. exactly the textbook RMSE. **These two rows therefore read the `rmse_textbook` key**, so the “RMSE” label is now literally correct rather than a convenient approximation.

Nothing is lost, and no cell is hand-typed. Both keys are written for every model, bank size and quantity, so a reader who wants the reproducibility report’s convention still has it in the JSON. The routing lives in `build_cross_tables.PS_METRIC_KEY` (which key each row reads) and `PS_METRIC_LABEL` (what each row is called), and both are read by the README renderer and by this document, so the two cannot disagree.

3 Why the difference exists

The repository originally used (A). Commit 99977a3, “Align strict path-shadowing RMSE with reproducibility report Table 5”, switched `_agg`, `_boot_ci` and the `terminal_rmse` diagnostic in all five scorers to (B) and re-ran the CSDI / LS4 / SBTS pipelines. Only the RMSE fields moved; every other metric reproduced bit-identically, and the Heston-oracle block stayed bit-identical across all three summaries. (The trajectory quantities have since moved back to (A) for the reason given in §2.4; both columns of the table below are read live, so it states the size of that move rather than assuming it.)

The measured inflation R_A/R_B at the 1M bank:

Quantity	Method	(A) root-last	(B) root-inside	ratio R_A/R_B
cum	CSDI	0.04925	0.04144	1.188
cum	LS4	0.04894	0.04145	1.181
cum	SBTS	0.05523	0.04674	1.182
step	CSDI	0.01245	0.01176	1.059
step	LS4	0.01245	0.01176	1.058
step	SBTS	0.01421	0.01350	1.053
rv	CSDI	0.01772	0.01373	1.291
rv	LS4	0.01819	0.01426	1.276
rv	SBTS	0.01972	0.01541	1.280

The ratio is near-constant *within* a quantity and very different *between* quantities (≈ 1.18 cum, ≈ 1.06 step, ≈ 1.28 rv). That is the signature predicted above: the gap is set by the dispersion of the per-path error distribution of the quantity, and across these three *generators* it barely moves.

One caveat, since it turned out to matter in §5. That stability holds across models of the *same kind*; it is not a law. Re-running the two conditional forecasters under (B) gave a cum ratio of 1.180 and a step ratio of 1.056, both in line with the generators — but an rv ratio of only 1.050, against the generators’ 1.28. Retrieval and direct forecasting produce differently dispersed per-path rv errors, and the ratio follows the dispersion. So the ratio may be used to *anticipate* the size of a correction, never to substitute for recomputing it.

4 Three defects in the reproducibility report

These are stated so the report can be corrected; none of them changes a number in the tables of Section 6.

1. §2.7 describes the opposite of the shipped code. The prose says “the per-path squared errors are averaged over the evaluation set first, and a *single* square root is taken last”, and the accompanying listing prints `np.sqrt(np.mean(per))` with the comment `# per holds se_i, NOT sqrt(se_i)`. The shipped code is `np.sqrt(per).mean()` — the reverse. The bootstrap listing is transposed in the same way. §2.7 therefore documents convention (A) while every table in the report except Table 8 is convention (B).

2. §2.7 states Jensen’s inequality backwards. It claims the mean-of-roots form “is $\geq$ the above whenever the se_i are not all equal, and would overstate RMSE”. Since $\sqrt{\cdot}$ is concave the inequality runs the other way: mean-of-roots *understates*. The data confirms the direction — every ratio in the table above exceeds 1.

3. The Table 8 caption swaps the two labels. It says Table 8’s column “was produced with an earlier aggregation convention (mean of per-path root errors)” and that Table 5 “follows the $\sqrt{\text{mean}_i(se_i)}$ definition”. Both halves are inverted: Table 8 is the root-last form and Table 5 is the mean-of-roots form. Table 8’s numbers are uniformly *larger*, which by Jensen is only possible if Table 8 is the root-last one. The caption on Table 5 (“RMSE fixed to $\text{mean}_i(\sqrt{se_i})$ ”) is the correct one, and it contradicts both §2.7 and the Table 8 caption.

Consequence. The repository was aligned against a *table caption* that contradicts the report’s own methods section. The alignment is self-consistent and reproducible, but the report should be corrected so a reader is not led to convention (A) by §2.7.

5 A mixed-convention defect in the tables — found and fixed

Until 2026-07-31 the RMSE rows of the five tables below mixed both conventions. The four generators, the Heston oracle and the RW floor used (B). The two conditional forecasters did not: `forecaster/pdf_bridge.py` pinned them to (A) through a local `metrics_frozen_rmse()`, so that a re-run would reproduce the reproducibility report’s published Chronos-2 / TimesFM cells rather than silently drift away from them.

Because $R_B \leq R_A$ always, this **systematically inflated the forecasters’ RMSE relative to every other column**. The defect was not cosmetic. `build_cross_tables.py::ps_winner_idx` ranks *all* model columns, forecasters included, so the Winner column was silently comparing inflated cells against un-inflated ones — while the README footnote next to it told the reader those cells were not comparable. A table cannot both rank a number and disclaim it.

The fix. `metrics_frozen_rmse()` was deleted; `score_forecaster` now calls the shared `P.metrics_with_ci`, and the `terminal_rmse` diagnostic was moved to `np.abs(...).mean()` to match `eval.bank` line 326 exactly. Both forecasters were then re-run from their seed-0 recipes. Every non-RMSE leaf of `chronos2_pdf.json` and `timesfm_pdf.json` reproduced bit-identically (83 and 85 leaves respectively; CRPS, coverages, widths and miss rates all unchanged to the last digit), which confirms the aggregation was the only thing that moved.

Method	Quantity	before (A)	after (B)	ratio
Chronos-2	cum	0.04921	0.04168	1.180
Chronos-2	step	0.01295	0.01227	1.056
Chronos-2	rv	0.23075	0.21983	1.050
TimesFM	cum	0.05094	0.04276	1.191
TimesFM	step	0.01332	0.01267	1.051
TimesFM	rv	0.28650	0.25696	1.115

What it changed. Under the common convention Chronos-2’s cum RMSE fell from 0.04921 to 0.04168, against CSDI’s 0.04144: a row that read as a 19% deficit became a 0.6% one. The estimate offered before the re-run — “roughly 0.0416” from the generator cum ratio — was accurate for cum and step, but the rv ratio came out at 1.050 rather than the generators’ 1.28 (see the caveat in §3), so only the recomputation settled it.

What it did not change. Win counts were recomputed at all five bank sizes and were identical: TimeDiT (raw) 14 at every bank, with CSDI 2 / SBTS 2 at 4096, CSDI 3 / SBTS 1 at 16384, and CSDI 3 / LS4 1 at the top three. CSDI remained the row minimum for cum and step RMSE, so **no Winner cell moved** — that statement is about *this* fix, the un-mixing of the forecaster column. The later switch of cum/step to the textbook aggregation *does* move Winner cells, and the paragraph after next measures by how much.

Why the tables in §6 now show the (A) column again. This is not a reversal. **The defect was the *mixing*, not the choice.** Once every column shared one convention it became possible to ask which one deserves the name “RMSE” — and for cum and step the answer is (A) (§2.4). So the trajectory rows moved to (A) *for all eight columns at once*: four generators, two forecasters, the Heston oracle and the RW floor, through the single PS_METRIC_KEY lookup. Comparability is preserved by construction; it is a property of the routing, not of a convention happening to be applied consistently by hand.

The ranking does *not* survive the move — and that is the real finding. It would be convenient to claim that a near-constant ratio R_A/R_B rescales each column monotonically and leaves the argmin alone. It does not. Recomputing every RMSE row under both aggregations, **7 of the 10 cum/step rows (2 quantities $\times$ 5 bank sizes) change winner**: CSDI takes every one of them under root-inside, and loses most of them to TimeDiT (raw) or LS4 under the textbook form.

That is not a defect in either convention. It is a measurement of how little separates the top three generators. At the 1M bank the cum RMSE values are 0.049253, 0.048875, 0.048944 for CSDI, TimeDiT (raw) and LS4 — a spread of 3.8×10^{-4} — while each 95% bootstrap CI is about 6.3×10^{-3} wide, $17\times$ larger. Every interval contains every other point estimate. **The cum and step RMSE winner among those three is noise**, and a reweighting as mild as moving one square root is enough to reshuffle it. Only SBTS (13% worse) and, more weakly, the two forecasters separate from the pack at all.

The practical consequence: read the RMSE rows of §6 as “CSDI $\approx$ TimeDiT (raw) $\approx$ LS4 < Chronos-2 < TimesFM $\ll$ SBTS”, and treat the bolded cell among the leading three as arbitrary. The CRPS, coverage and miss rows are the ones that discriminate. The win counts printed under each table are recomputed from the shipped artefacts every build, never assumed.

The cost. The reproducibility report’s published Chronos-2 / TimesFM *rv* cell is now *superseded, not reproduced*: *rv* reports (B) and is labelled MAE, so a re-run will not match the report’s *rv* column, by design. The cum and step cells, having moved to (A), agree with the report’s Table 8 convention again. That is the correct trade: one internally consistent table, with both aggregations preserved in the JSON for anyone who needs the other. This is recorded in GUIDELINE E16b.

6 The five cross-method tables

Each table is one nested-prefix bank size of the *same* 1M bank (GUIDELINE §9.1). Generator and oracle columns move with bank size; the forecaster columns and the RW floor have no bank-size axis and repeat unchanged. Winner excludes the oracle and RW floors. Width rows are diagnostics and carry no winner: a degenerate zero-width interval would “win” while being completely uncalibrated. **Bold** = best in row. TimeDiT’s column is labelled **TimeDiT (raw)** because its bank comes from the no-preprocessing checkpoint; the protocol, $K = 256$, query set, embedding, oracle and floor are identical to the other generators.

Every number below is generated by the same code path as the README tables (`build_cross_tables.py`), reading the same `pdf_summary.json` files, so the two cannot drift apart.

6.1 Bank size $N = 4096$

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04919	0.04890	0.04906	0.04916	0.04921	0.05094	0.04924	0.05670	TimeDiT (raw)
CRPS ↓	0.02548	0.02531	0.02544	0.02544	0.02624	0.02722	0.02539	0.02946	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4571	0.5105	0.4764	0.4839	0.4050	0.4664	0.4875	0.4719	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8629	0.8957	0.8691	0.8805	0.7905	0.8177	0.8919	0.8553	TimeDiT (raw)
width ₅₀ (diag)	0.05070	0.05901	0.05376	0.05536	0.04454	0.05477	0.05573	0.06286	—
width ₉₀ (diag)	0.1318	0.1488	0.1353	0.1439	0.1166	0.1348	0.1462	0.1520	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07764	0.04840	0.06476	0.05554	0.09595	0.09265	0.05731	0.08331	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.05945	0.05591	0.06616	0.06396	0.1135	0.08966	0.05078	0.06134	TimeDiT (raw)
step ($M \times H$)									
RMSE ↓	0.01244	0.01245	0.01245	0.01249	0.01295	0.01332	0.01245	0.01253	CSDI
CRPS ↓	0.006780	0.006766	0.006794	0.006812	0.01272	0.01473	0.006765	0.006874	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4510	0.4811	0.4656	0.4813	0.9449	0.9479	0.4849	0.5157	SBTS
coverage ₉₀ ($\rightarrow 0.90$)	0.8602	0.8854	0.8591	0.8782	0.9970	0.9949	0.8869	0.8866	TimeDiT (raw)
width ₅₀ (diag)	0.01321	0.01429	0.01359	0.01434	0.06405	0.07667	0.01434	0.01602	—
width ₉₀ (diag)	0.03524	0.03794	0.03530	0.03769	0.1609	0.1801	0.03828	0.03968	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07098	0.05573	0.07233	0.05890	0.001770	0.002319	0.05676	0.05933	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.06885	0.05884	0.06860	0.06287	0.001221	0.002808	0.05634	0.05408	TimeDiT (raw)
rv (M scalar)									
MAE ↓	0.01458	0.01364	0.01516	0.01420	0.2198	0.2570	0.01381	0.01509	TimeDiT (raw)
CRPS ↓	0.01055	0.009653	0.01098	0.01010	0.1913	0.2291	0.009777	0.01168	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4609	0.4805	0.4121	0.4805	0	0	0.5059	0.2344	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8594	0.8945	0.7812	0.8652	0	0.001953	0.9141	0.5332	TimeDiT (raw)
width ₅₀ (diag)	0.02220	0.02273	0.02027	0.02287	0.06861	0.06689	0.02404	0.01202	—
width ₉₀ (diag)	0.05413	0.05489	0.04837	0.05458	0.1651	0.1610	0.05753	0.02873	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.01758	0.04102	0.05664	0.05078	1.000	0.9980	0.02734	0.2891	SBTS
upper miss ₉₀ ($\rightarrow 0.05$)	0.1230	0.06445	0.1621	0.08398	0	0	0.05859	0.1777	TimeDiT (raw)

Winners (of 18 ranked rows = 6 ranked metrics $\times$ 3 quantities): TimeDiT (raw) 15 · SBTS 2 · CSDI 1.

6.2 Bank size $N = 16\,384$

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04915	0.04901	0.04903	0.04996	0.04921	0.05094	0.04923	0.05670	TimeDiT (raw)
CRPS ↓	0.02543	0.02534	0.02542	0.02610	0.02624	0.02722	0.02535	0.02946	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4604	0.5103	0.4755	0.4694	0.4050	0.4664	0.4859	0.4719	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8694	0.8952	0.8685	0.8644	0.7905	0.8177	0.8884	0.8553	TimeDiT (raw)
width ₅₀ (diag)	0.05197	0.05936	0.05375	0.05538	0.04454	0.05477	0.05606	0.06286	—
width ₉₀ (diag)	0.1348	0.1484	0.1357	0.1410	0.1166	0.1348	0.1446	0.1520	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07318	0.04651	0.06458	0.06787	0.09595	0.09265	0.05731	0.08331	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.05743	0.05829	0.06689	0.06769	0.1135	0.08966	0.05432	0.06134	CSDI
step ($M \times H$)									
RMSE ↓	0.01245	0.01245	0.01245	0.01261	0.01295	0.01332	0.01245	0.01253	LS4
CRPS ↓	0.006777	0.006768	0.006788	0.006930	0.01272	0.01473	0.006761	0.006874	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4493	0.4818	0.4648	0.4751	0.9449	0.9479	0.4810	0.5157	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8639	0.8843	0.8630	0.8624	0.9970	0.9949	0.8862	0.8866	TimeDiT (raw)
width ₅₀ (diag)	0.01326	0.01431	0.01365	0.01444	0.06405	0.07667	0.01437	0.01602	—
width ₉₀ (diag)	0.03534	0.03781	0.03544	0.03709	0.1609	0.1801	0.03807	0.03968	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07104	0.05719	0.06995	0.06750	0.001770	0.002319	0.05762	0.05933	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.06506	0.05847	0.06702	0.07013	0.001221	0.002808	0.05621	0.05408	TimeDiT (raw)
rv (M scalar)									
MAE ↓	0.01421	0.01341	0.01470	0.01414	0.2198	0.2570	0.01347	0.01509	TimeDiT (raw)
CRPS ↓	0.01028	0.009484	0.01066	0.01009	0.1913	0.2291	0.009513	0.01168	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4570	0.4883	0.4199	0.4629	0	0	0.4961	0.2344	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8691	0.8926	0.7871	0.8574	0	0.001953	0.9141	0.5332	TimeDiT (raw)
width ₅₀ (diag)	0.02189	0.02217	0.01990	0.02194	0.06861	0.06689	0.02286	0.01202	—
width ₉₀ (diag)	0.05318	0.05322	0.04786	0.05252	0.1651	0.1610	0.05559	0.02873	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.01562	0.03906	0.05469	0.05273	1.000	0.9980	0.03125	0.2891	SBTS
upper miss ₉₀ ($\rightarrow 0.05$)	0.1152	0.06836	0.1582	0.08984	0	0	0.05469	0.1777	TimeDiT (raw)

Winners (of 18 ranked rows = 6 ranked metrics $\times$ 3 quantities): TimeDiT (raw) 15 · CSDI 1 · LS4 1 · SBTS 1.

6.3 Bank size $N = 65\,536$

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04912	0.04909	0.04900	0.05207	0.04921	0.05094	0.04910	0.05670	LS4
CRPS ↓	0.02535	0.02533	0.02536	0.02768	0.02624	0.02722	0.02524	0.02946	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4670	0.5046	0.4717	0.4256	0.4050	0.4664	0.4886	0.4719	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8738	0.8970	0.8685	0.8198	0.7905	0.8177	0.8887	0.8553	TimeDiT (raw)
width ₅₀ (diag)	0.05283	0.05886	0.05369	0.05231	0.04454	0.05477	0.05630	0.06286	—
width ₉₀ (diag)	0.1354	0.1480	0.1353	0.1309	0.1166	0.1348	0.1448	0.1520	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07147	0.04761	0.06500	0.08795	0.09595	0.09265	0.05688	0.08331	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.05469	0.05536	0.06647	0.09229	0.1135	0.08966	0.05444	0.06134	CSDI
step ($M \times H$)									
RMSE ↓	0.01244	0.01245	0.01246	0.01297	0.01295	0.01332	0.01245	0.01253	CSDI
CRPS ↓	0.006770	0.006762	0.006786	0.007304	0.01272	0.01473	0.006754	0.006874	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4486	0.4765	0.4667	0.4406	0.9449	0.9479	0.4820	0.5157	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8639	0.8810	0.8627	0.8183	0.9970	0.9949	0.8864	0.8866	TimeDiT (raw)
width ₅₀ (diag)	0.01330	0.01423	0.01367	0.01400	0.06405	0.07667	0.01441	0.01602	—
width ₉₀ (diag)	0.03542	0.03760	0.03538	0.03433	0.1609	0.1801	0.03807	0.03968	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07001	0.05884	0.07025	0.08905	0.001770	0.002319	0.05658	0.05933	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.06610	0.06012	0.06708	0.09265	0.001221	0.002808	0.05701	0.05408	TimeDiT (raw)
rv (M scalar)									
MAE ↓	0.01393	0.01327	0.01448	0.01432	0.2198	0.2570	0.01314	0.01509	TimeDiT (raw)
CRPS ↓	0.01007	0.009409	0.01048	0.01058	0.1913	0.2291	0.009330	0.01168	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4551	0.4902	0.4141	0.4258	0	0	0.4980	0.2344	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8652	0.8945	0.7910	0.8262	0	0.001953	0.9199	0.5332	TimeDiT (raw)
width ₅₀ (diag)	0.02148	0.02197	0.01961	0.02067	0.06861	0.06689	0.02259	0.01202	—
width ₉₀ (diag)	0.05219	0.05302	0.04666	0.04806	0.1651	0.1610	0.05502	0.02873	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.01562	0.03711	0.05078	0.06445	1.000	0.9980	0.02930	0.2891	LS4
upper miss ₉₀ ($\rightarrow 0.05$)	0.1191	0.06836	0.1582	0.1094	0	0	0.05078	0.1777	TimeDiT (raw)

Winners (of 18 ranked rows = 6 ranked metrics $\times$ 3 quantities): TimeDiT (raw) 14 · CSDI 2 · LS4 2.

6.4 Bank size $N = 262\,144$

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04911	0.04891	0.04897	0.05373	0.04921	0.05094	0.04898	0.05670	TimeDiT (raw)
CRPS ↓	0.02541	0.02528	0.02536	0.02987	0.02624	0.02722	0.02524	0.02946	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4644	0.5010	0.4723	0.3552	0.4050	0.4664	0.4904	0.4719	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8691	0.8972	0.8696	0.7396	0.7905	0.8177	0.8922	0.8553	TimeDiT (raw)
width ₅₀ (diag)	0.05267	0.05879	0.05349	0.04573	0.04454	0.05477	0.05706	0.06286	—
width ₉₀ (diag)	0.1344	0.1480	0.1350	0.1137	0.1166	0.1348	0.1460	0.1520	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07501	0.04675	0.06653	0.1324	0.09595	0.09265	0.05548	0.08331	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.05591	0.05609	0.06384	0.1280	0.1135	0.08966	0.05231	0.06134	CSDI
step ($M \times H$)									
RMSE ↓	0.01245	0.01245	0.01245	0.01360	0.01295	0.01332	0.01245	0.01253	CSDI
CRPS ↓	0.006771	0.006760	0.006784	0.008009	0.01272	0.01473	0.006752	0.006874	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4529	0.4753	0.4653	0.3699	0.9449	0.9479	0.4825	0.5157	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8638	0.8622	0.8645	0.7321	0.9970	0.9949	0.8849	0.8866	TimeDiT (raw)
width ₅₀ (diag)	0.01332	0.01425	0.01367	0.01230	0.06405	0.07667	0.01444	0.01602	—
width ₉₀ (diag)	0.03547	0.03759	0.03538	0.02975	0.1609	0.1801	0.03806	0.03968	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.07080	0.05811	0.06830	0.1332	0.001770	0.002319	0.05829	0.05933	TimeDiT (raw)
upper miss ₉₀ ($\rightarrow 0.05$)	0.06543	0.05969	0.06720	0.1347	0.001221	0.002808	0.05682	0.05408	TimeDiT (raw)
rv (M scalar)									
MAE ↓	0.01379	0.01305	0.01433	0.01487	0.2198	0.2570	0.01306	0.01509	TimeDiT (raw)
CRPS ↓	0.009987	0.009273	0.01038	0.01155	0.1913	0.2291	0.009256	0.01168	TimeDiT (raw)
coverage ₅₀ ($\rightarrow 0.50$)	0.4609	0.4902	0.4277	0.3496	0	0	0.4922	0.2344	TimeDiT (raw)
coverage ₉₀ ($\rightarrow 0.90$)	0.8672	0.8926	0.8105	0.7500	0	0.001953	0.9180	0.5332	TimeDiT (raw)
width ₅₀ (diag)	0.02140	0.02177	0.01929	0.01770	0.06861	0.06689	0.02231	0.01202	—
width ₉₀ (diag)	0.05196	0.05261	0.04621	0.04149	0.1651	0.1610	0.05430	0.02873	—
lower miss ₉₀ ($\rightarrow 0.05$)	0.01758	0.03906	0.04688	0.09766	1.000	0.9980	0.03125	0.2891	LS4
upper miss ₉₀ ($\rightarrow 0.05$)	0.1152	0.06836	0.1426	0.1523	0	0	0.05078	0.1777	TimeDiT (raw)

Winners (of 18 ranked rows = 6 ranked metrics $\times$ 3 quantities): TimeDiT (raw) 15 · CSDI 2 · LS4 1.

6.5 Bank size $N = 1\,000\,000$

Quantity / metric	CSDI	TimeDiT (raw)	LS4	SBTS	Chronos-2	TimesFM	Oracle	RW	Winner
cum ($M \times H$ trajectory)									
RMSE ↓	0.04925	0.04888	0.04894	0.05523	0.04921	0.05094	0.04911	0.05670	TimeDiT (raw)
CRPS ↓	0.02544	0.02521	0.02534	0.03179	0.02624	0.02722	0.02525	0.02946	TimeDiT (raw)
coverage ₅₀ (→0.50)	0.4624	0.5072	0.4688	0.3001	0.4050	0.4664	0.4891	0.4719	TimeDiT (raw)
coverage ₉₀ (→0.90)	0.8682	0.8983	0.8682	0.6575	0.7905	0.8177	0.8941	0.8553	TimeDiT (raw)
width ₅₀ (diag)	0.05285	0.05901	0.05359	0.04003	0.04454	0.05477	0.05719	0.06286	—
width ₉₀ (diag)	0.1348	0.1486	0.1349	0.09761	0.1166	0.1348	0.1465	0.1520	—
lower miss ₉₀ (→0.05)	0.07599	0.04572	0.06757	0.1738	0.09595	0.09265	0.05463	0.08331	TimeDiT (raw)
upper miss ₉₀ (→0.05)	0.05579	0.05597	0.06427	0.1687	0.1135	0.08966	0.05127	0.06134	CSDI
step ($M \times H$)									
RMSE ↓	0.01245	0.01246	0.01245	0.01421	0.01295	0.01332	0.01244	0.01253	LS4
CRPS ↓	0.006767	0.006760	0.006778	0.008688	0.01272	0.01473	0.006750	0.006874	TimeDiT (raw)
coverage ₅₀ (→0.50)	0.4532	0.4761	0.4672	0.3082	0.9449	0.9479	0.4816	0.5157	TimeDiT (raw)
coverage ₉₀ (→0.90)	0.8640	0.8806	0.8640	0.6479	0.9970	0.9949	0.8864	0.8866	TimeDiT (raw)
width ₅₀ (diag)	0.01338	0.01428	0.01368	0.01071	0.06405	0.07667	0.01449	0.01602	—
width ₉₀ (diag)	0.03550	0.03758	0.03538	0.02568	0.1609	0.1801	0.03815	0.03968	—
lower miss ₉₀ (→0.05)	0.07062	0.05884	0.06921	0.1762	0.001770	0.002319	0.05725	0.05933	TimeDiT (raw)
upper miss ₉₀ (→0.05)	0.06543	0.06061	0.06683	0.1758	0.001221	0.002808	0.05634	0.05408	TimeDiT (raw)
rv (M scalar)									
MAE ↓	0.01373	0.01295	0.01426	0.01541	0.2198	0.2570	0.01291	0.01509	TimeDiT (raw)
CRPS ↓	0.009938	0.009179	0.01032	0.01241	0.1913	0.2291	0.009143	0.01168	TimeDiT (raw)
coverage ₅₀ (→0.50)	0.4746	0.4883	0.4102	0.2930	0	0	0.4727	0.2344	TimeDiT (raw)
coverage ₉₀ (→0.90)	0.8633	0.8945	0.7988	0.6641	0	0.001953	0.9238	0.5332	TimeDiT (raw)
width ₅₀ (diag)	0.02115	0.02155	0.01914	0.01493	0.06861	0.06689	0.02217	0.01202	—
width ₉₀ (diag)	0.05138	0.05213	0.04589	0.03608	0.1651	0.1610	0.05411	0.02873	—
lower miss ₉₀ (→0.05)	0.01758	0.03125	0.05078	0.1328	1.000	0.9980	0.02930	0.2891	LS4
upper miss ₉₀ (→0.05)	0.1191	0.07422	0.1504	0.2031	0	0	0.04688	0.1777	TimeDiT (raw)

Winners (of 18 ranked rows = 6 ranked metrics × 3 quantities): TimeDiT (raw) 15 · LS4 2 · CSDI 1.

7 Figures

Every figure in the experiment folder, grouped. The eight-panel diagnostics are, row by row: real vs generated path bundles; log-return density and QQ vs real; ACF of $|r|$ and ACF of r^2 ; 5-day rolling-volatility density and terminal-price tail survival (log-log). Blue = real test set, red/orange = generated, dashed = theoretical or empirical reference.

Known caption artefact: the path-bundle panel titles read “8192 total”; the true count is 4096. The shared plotter `metrics/plot_diagnostics.py` hardcodes 8192 (lines 151/160) from the main benchmark. The plotted data is the correct 4096-path test set — only the string is stale, it affects all four methods identically, and it changes no number in any table.

7.1 Stylised-fact diagnostics (8-panel)

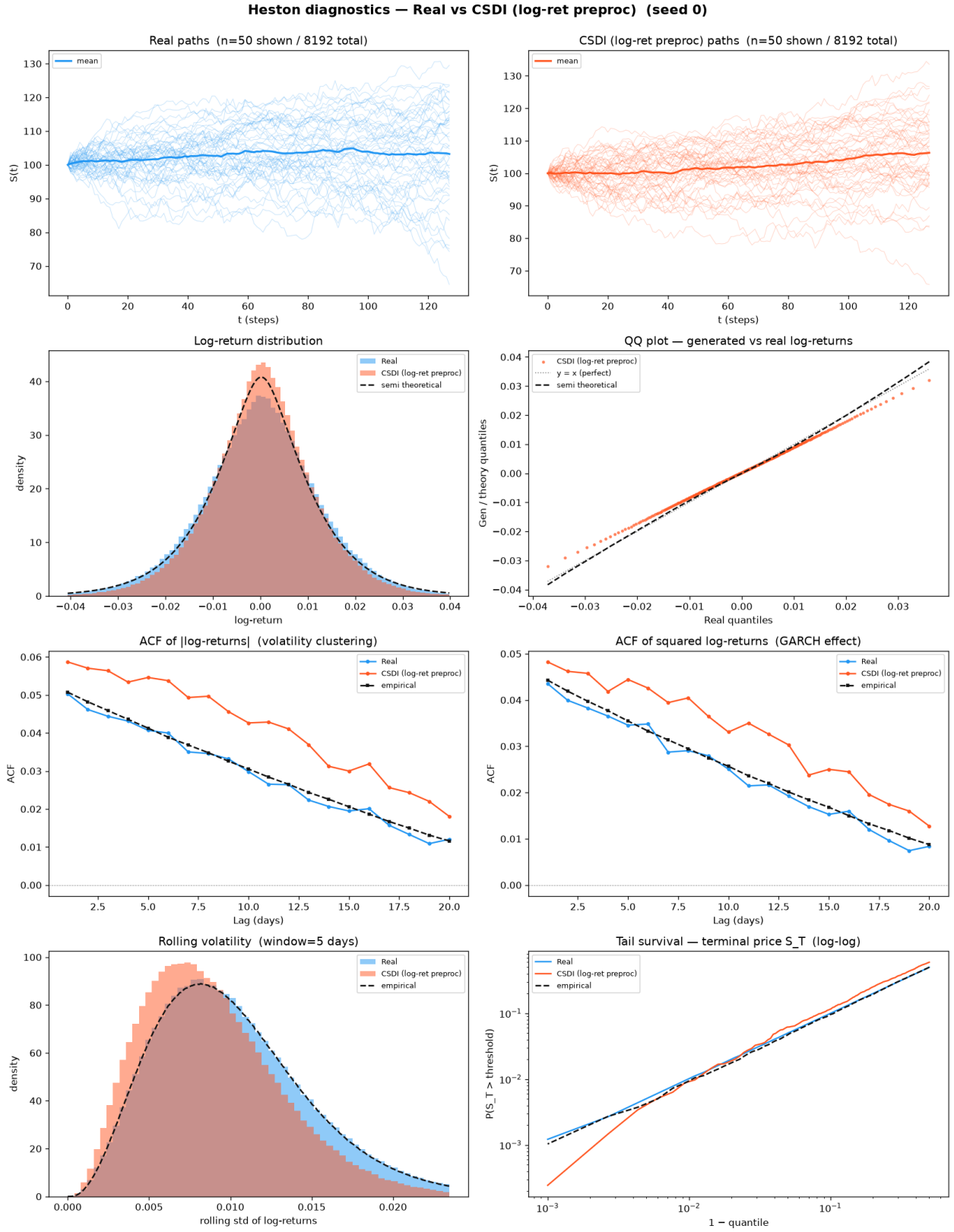


Figure 1: CSDI — log-return preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

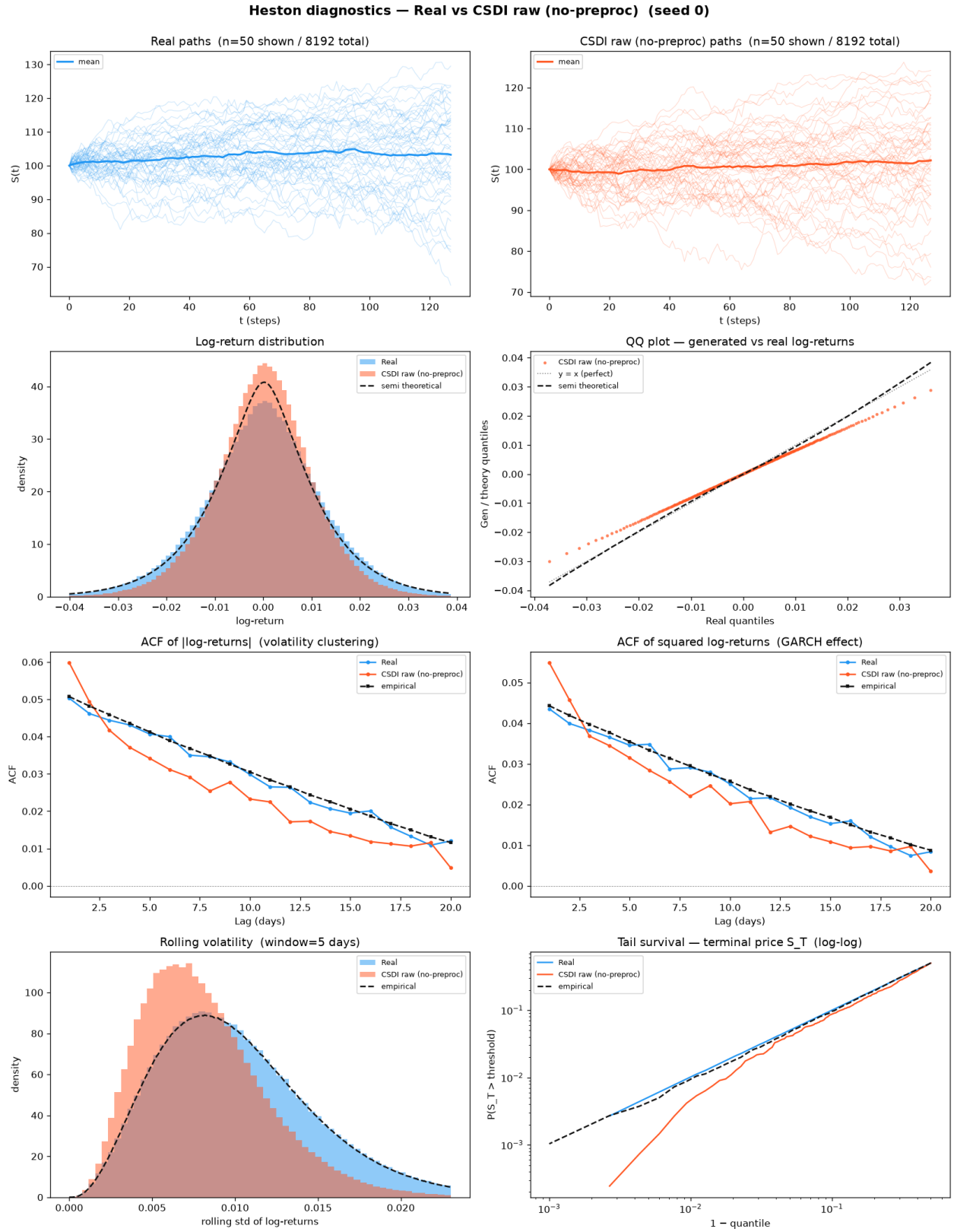


Figure 2: CSDI — raw price, no preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

Heston diagnostics — Real vs TimeDiT (log-ret preproc) (seed 0)

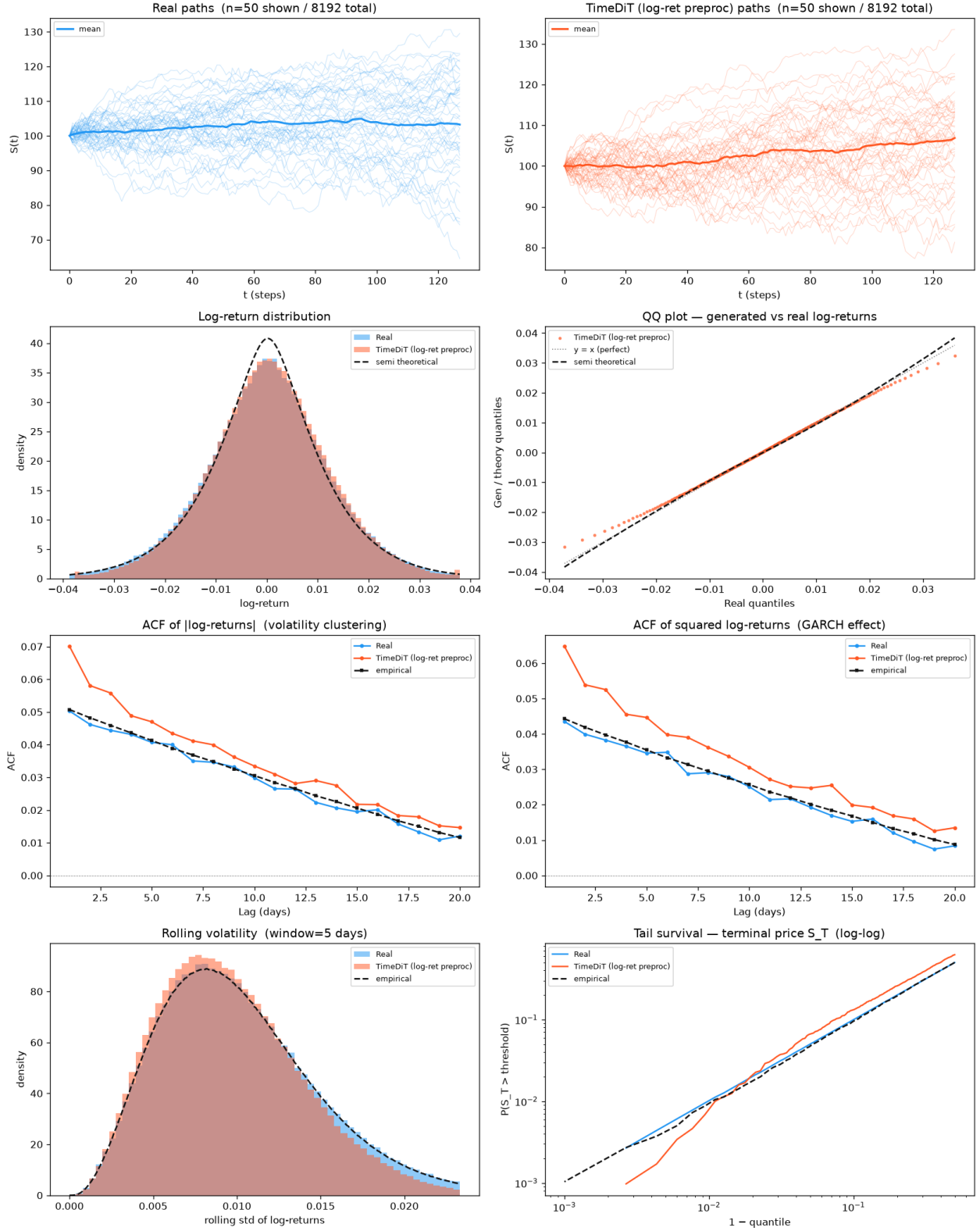


Figure 3: TimeDiT — log-return preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

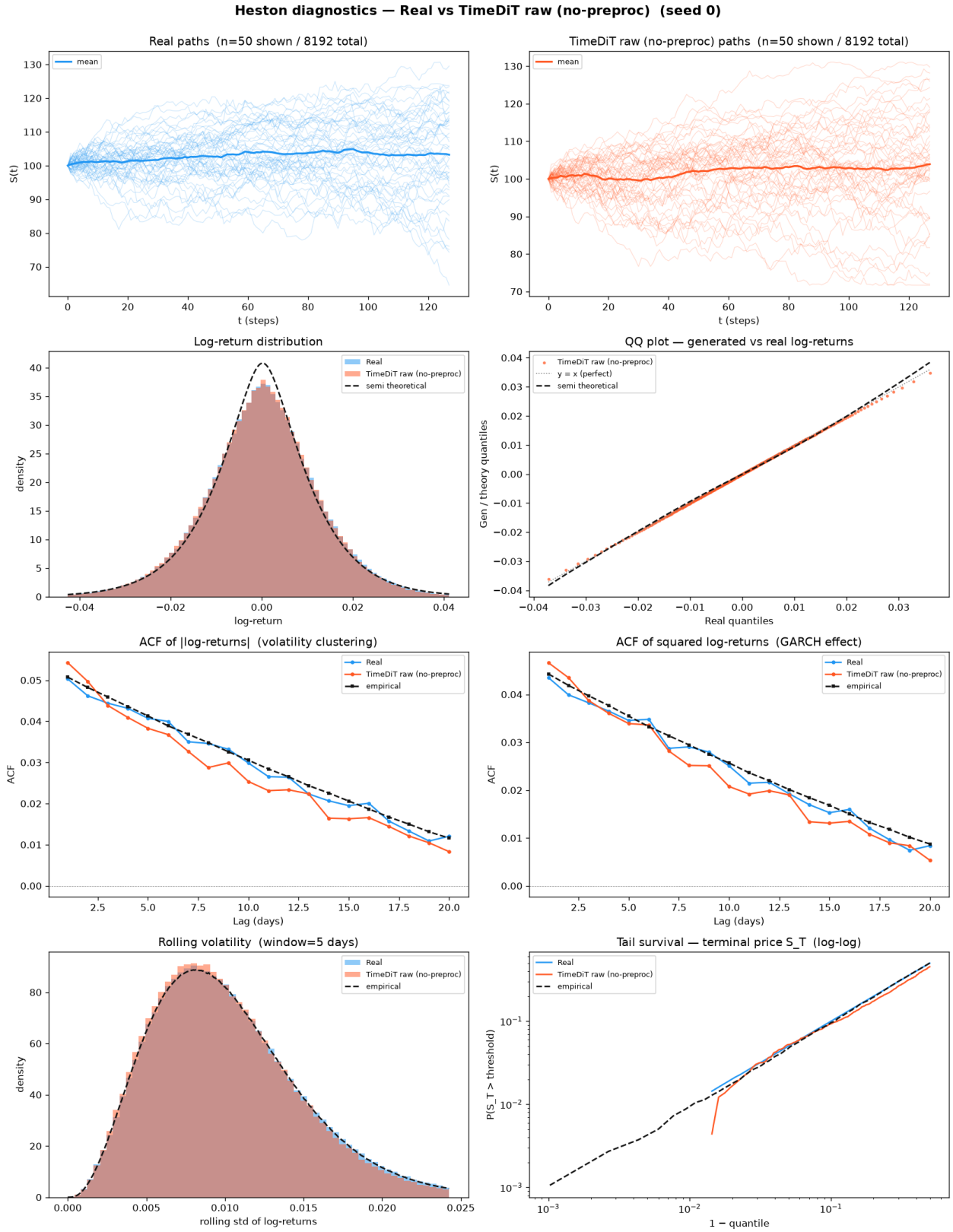


Figure 4: TimeDiT — raw price, no preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

Heston diagnostics — Real vs LS4 (log-ret preproc) (seed 0)

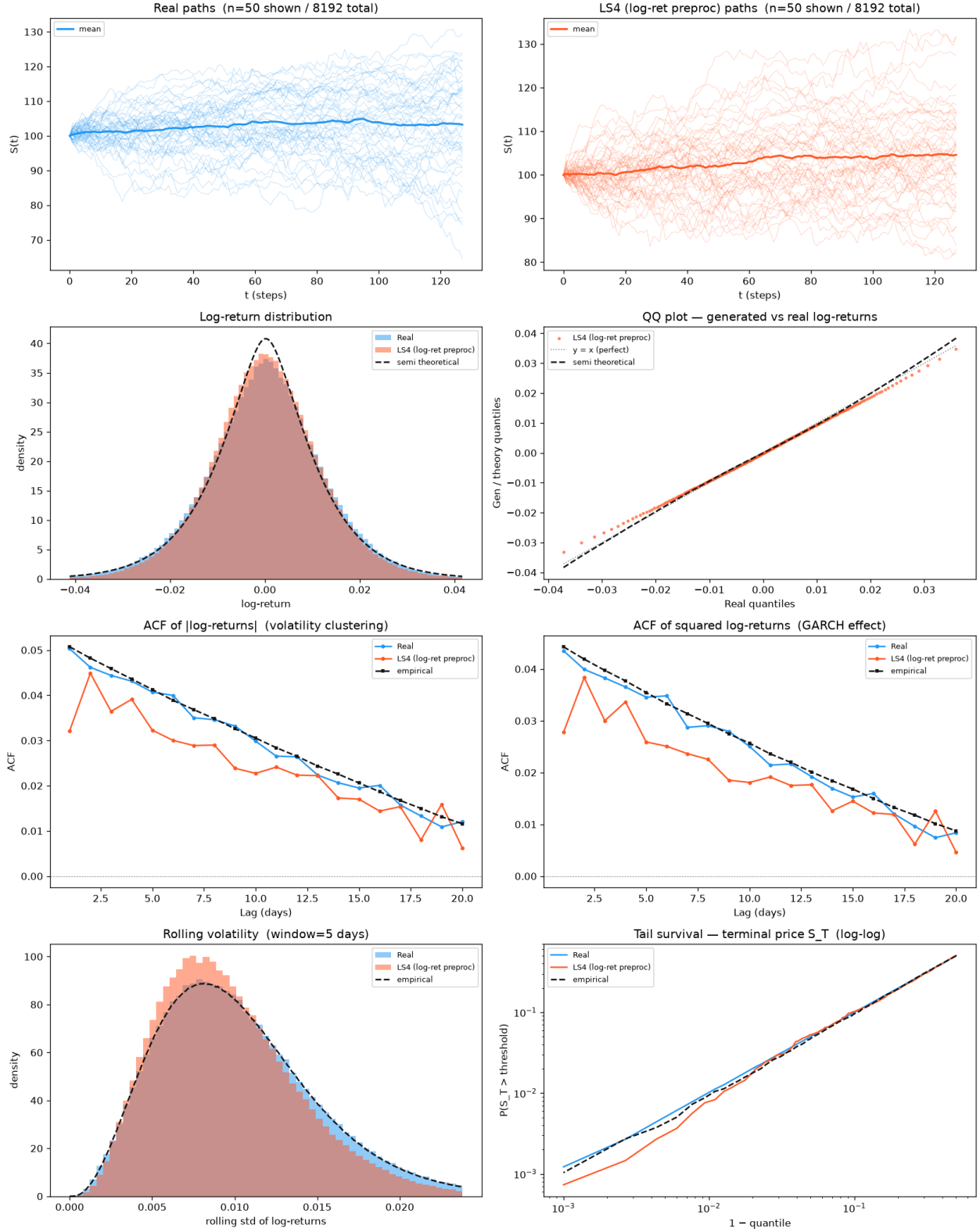


Figure 5: LS4 — log-return preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

Heston diagnostics — Real vs LS4 raw (no-preproc) (seed 0)

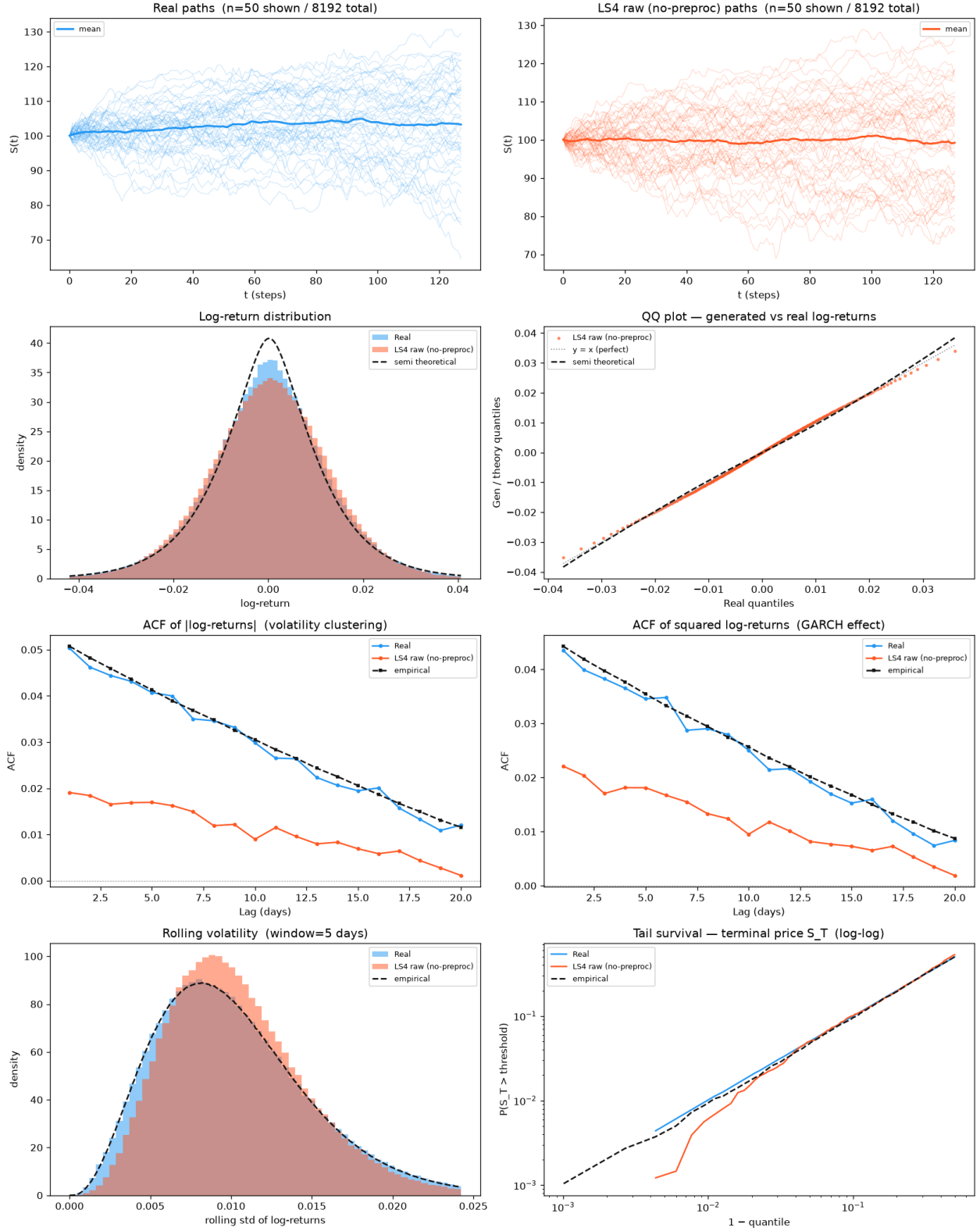


Figure 6: LS4 — raw price, no preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

Heston diagnostics — Real vs SBTS (log-ret preproc) (seed 0)

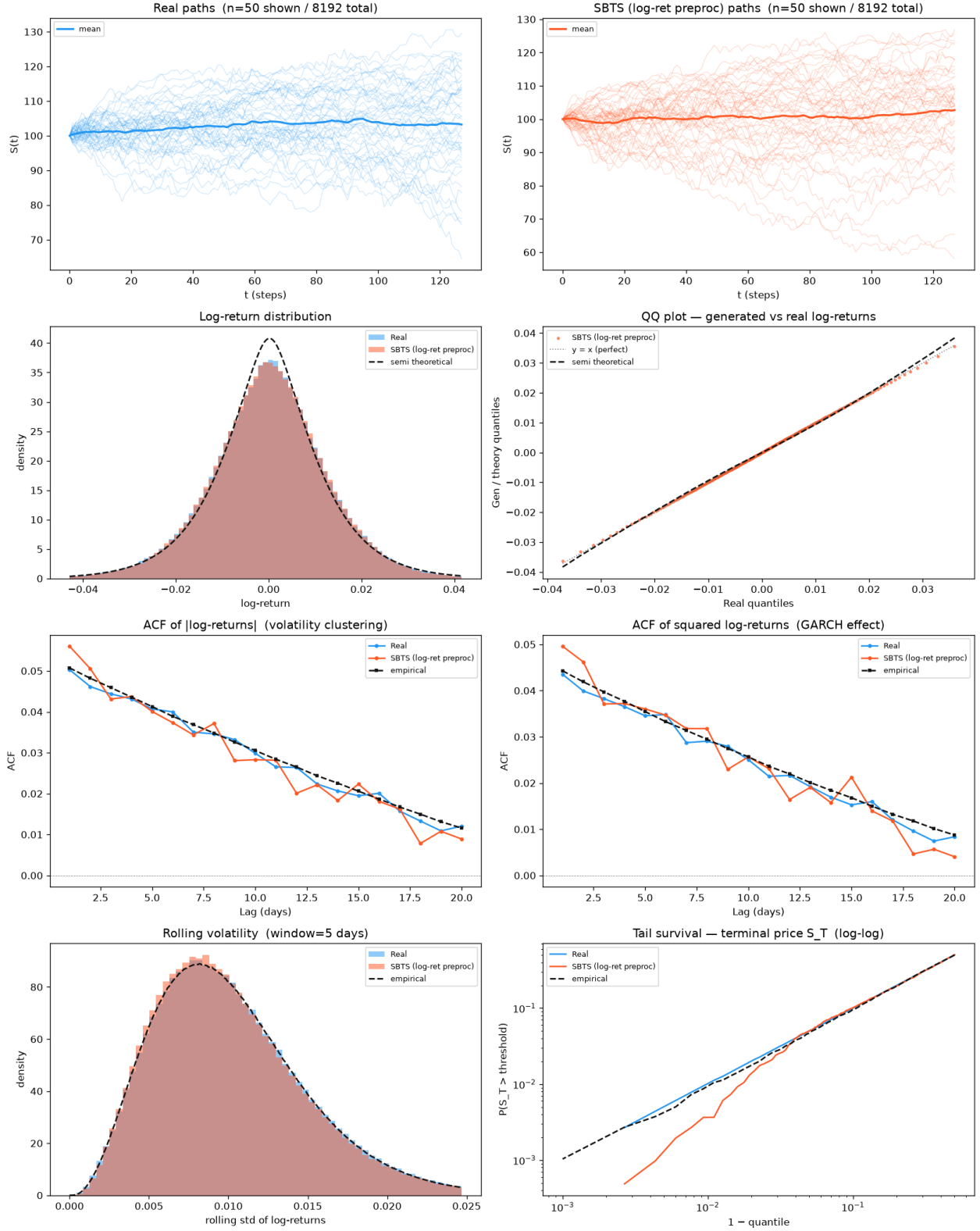


Figure 7: SBTS — log-return preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

Heston diagnostics — Real vs SBTS (no preproc, raw-price kernel) (seed 0)

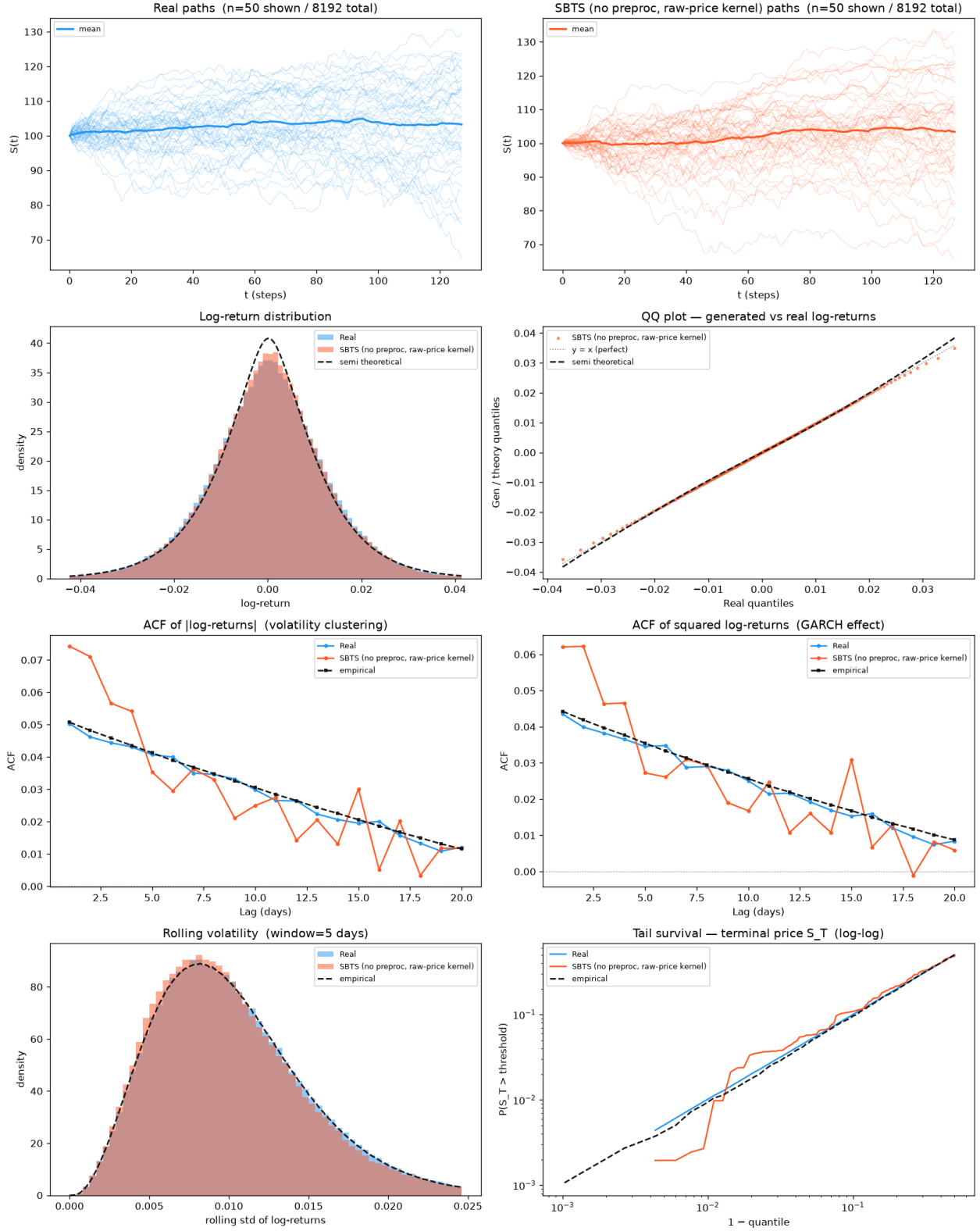


Figure 8: SBTS — raw price, no preprocessing. Eight-panel Heston stylised-fact diagnostics (seed 0).

7.2 Path-shadowing convergence and calibration

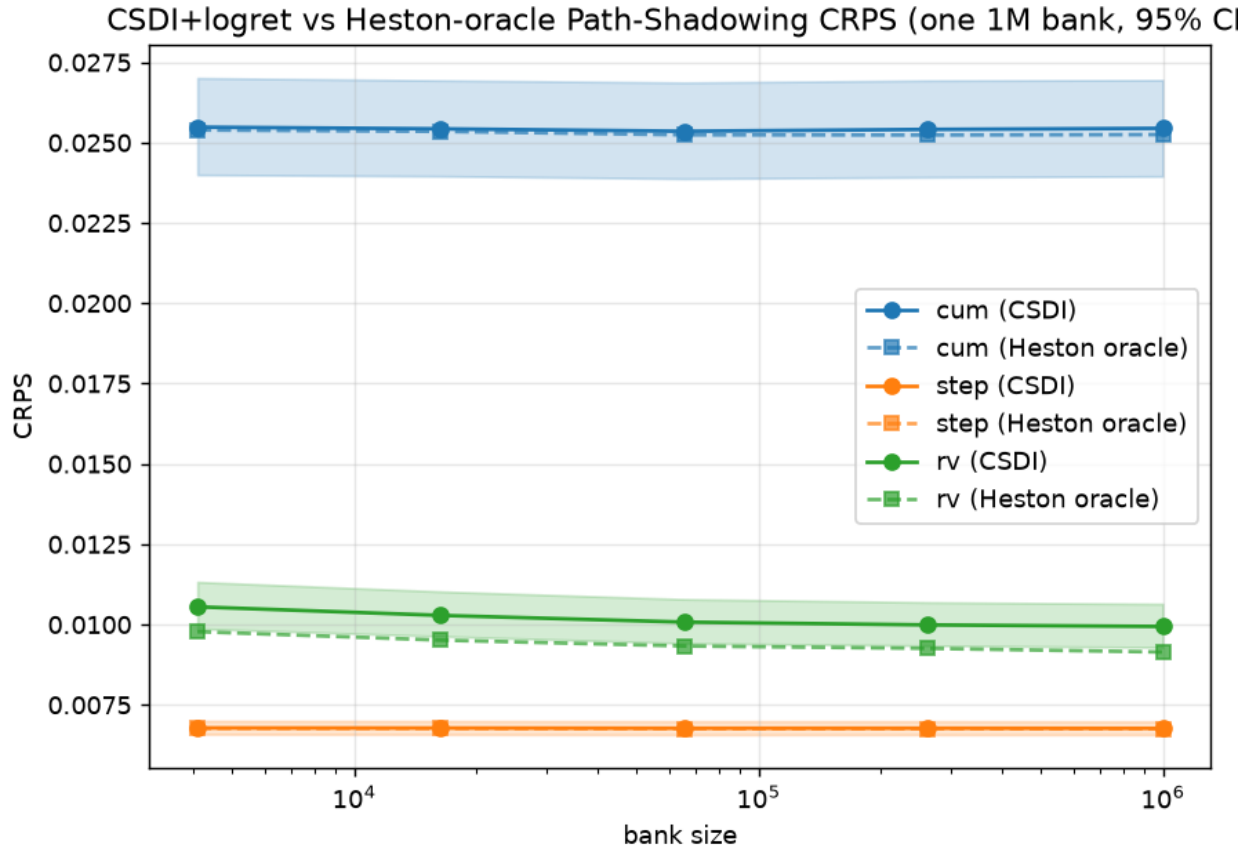


Figure 9: CSDI — CRPS vs bank size ($\log x$); solid = generator, dashed = size-matched Heston oracle, shaded = 95% bootstrap band.

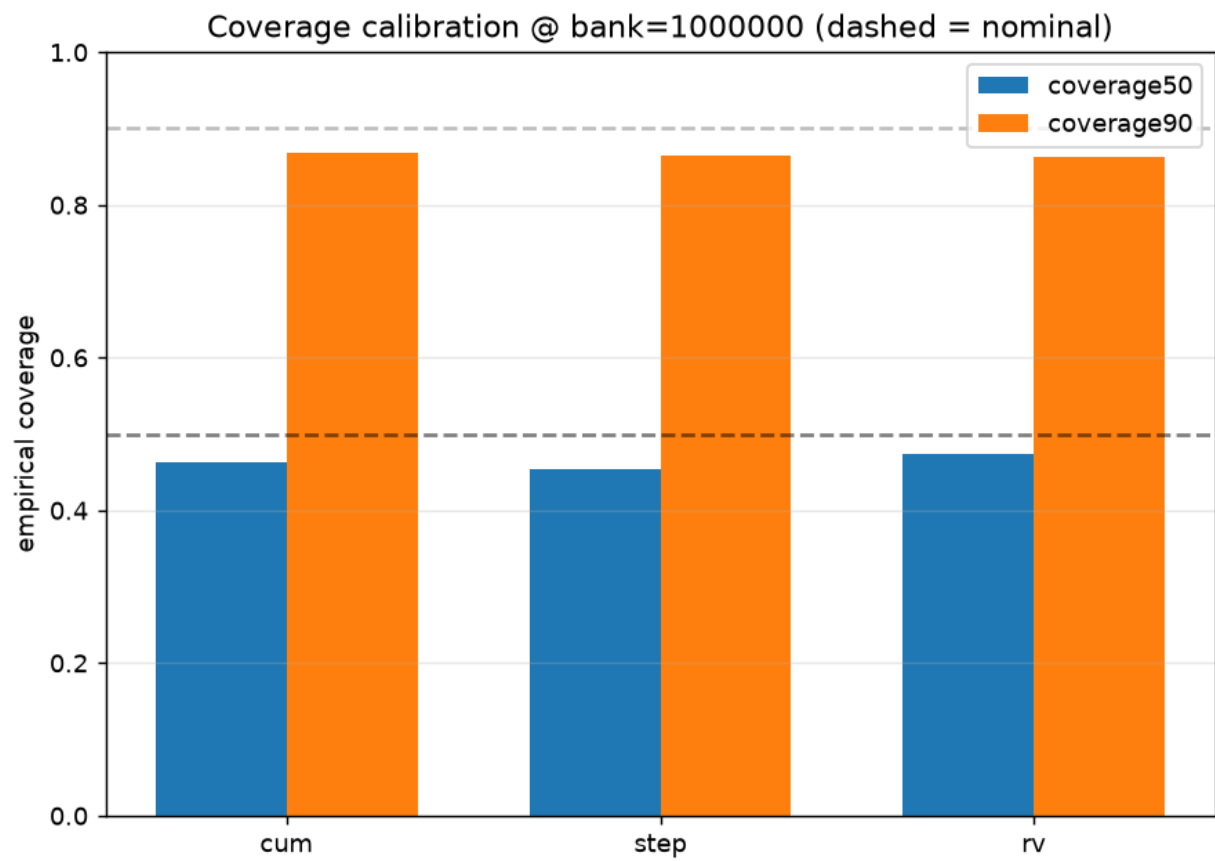


Figure 10: CSDI — Coverage calibration at the 1M bank; dashed lines = 0.50 / 0.90 nominal targets.

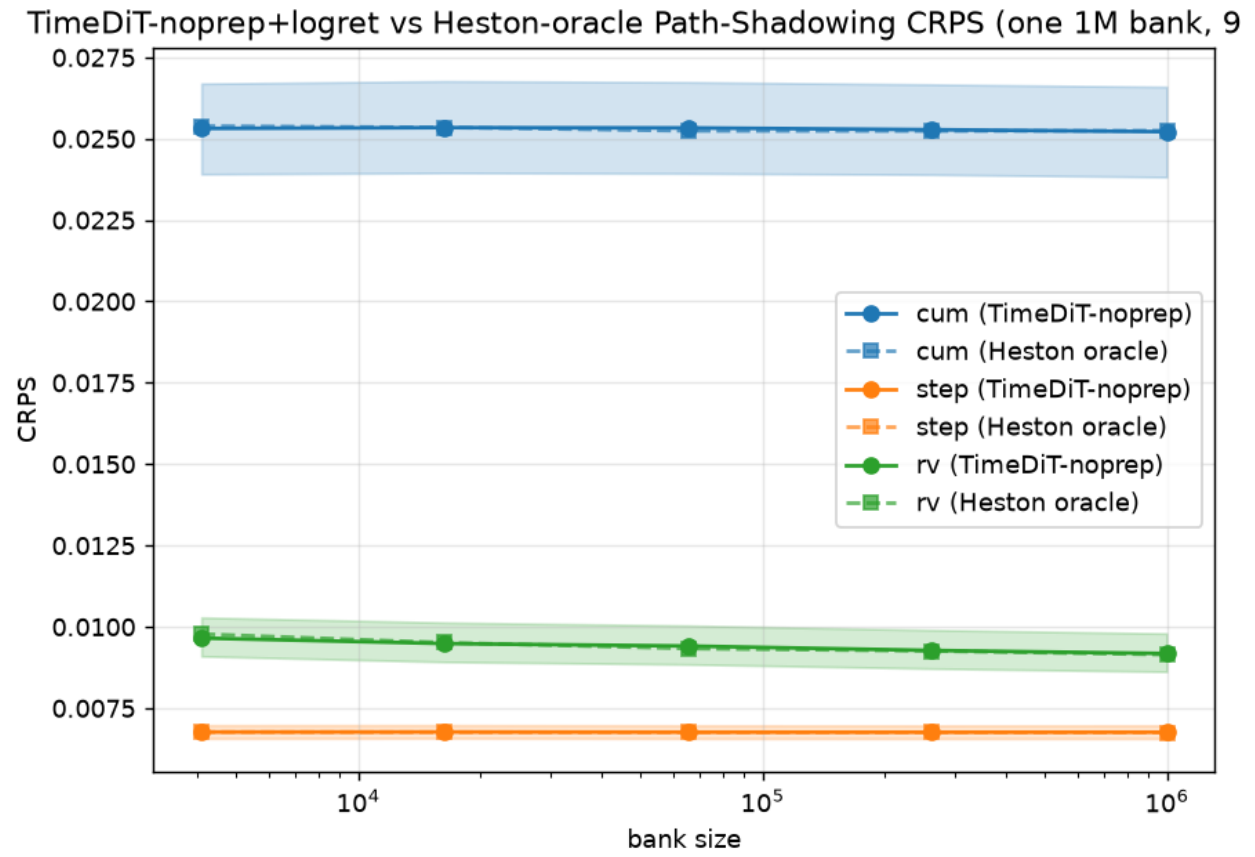


Figure 11: TimeDiT (raw) — CRPS vs bank size ($\log x$); solid = generator, dashed = size-matched Heston oracle, shaded = 95% bootstrap band.

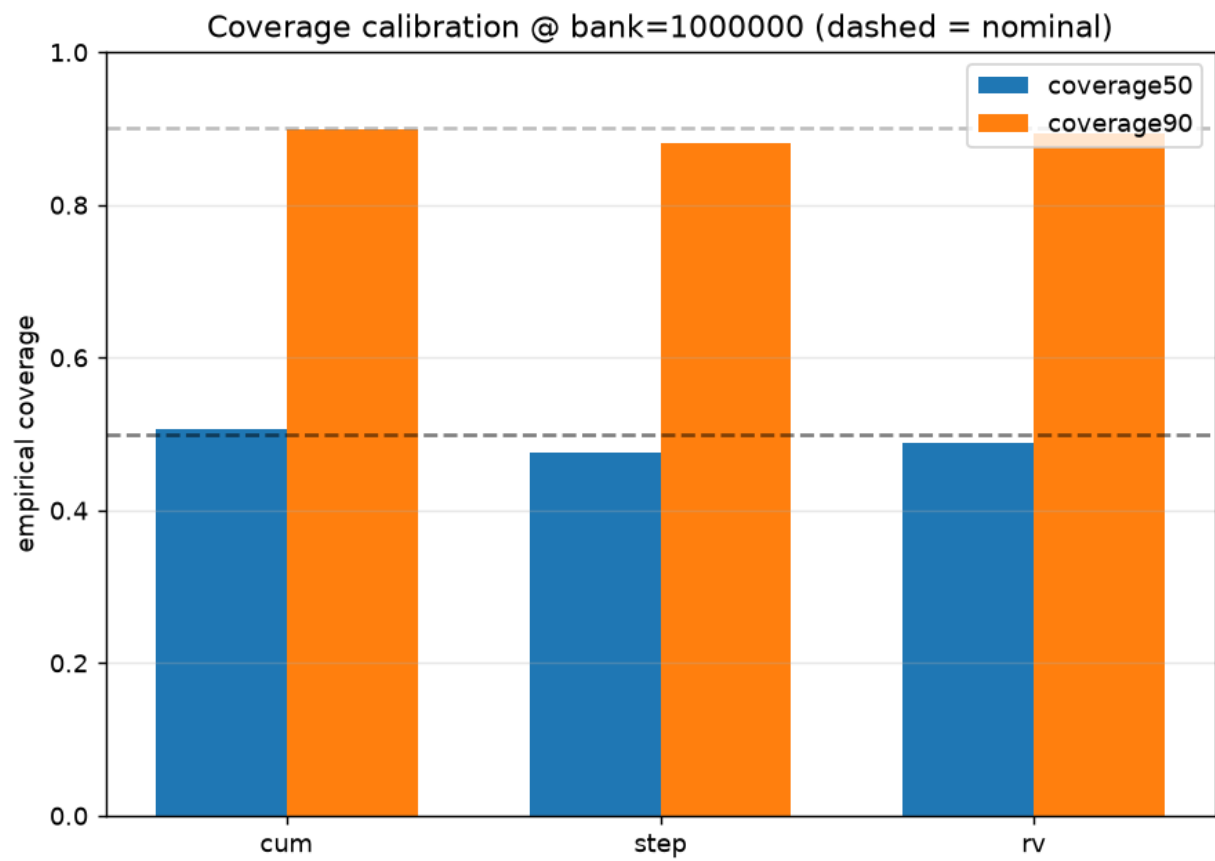


Figure 12: TimeDiT (raw) — Coverage calibration at the 1M bank; dashed lines = 0.50 / 0.90 nominal targets.

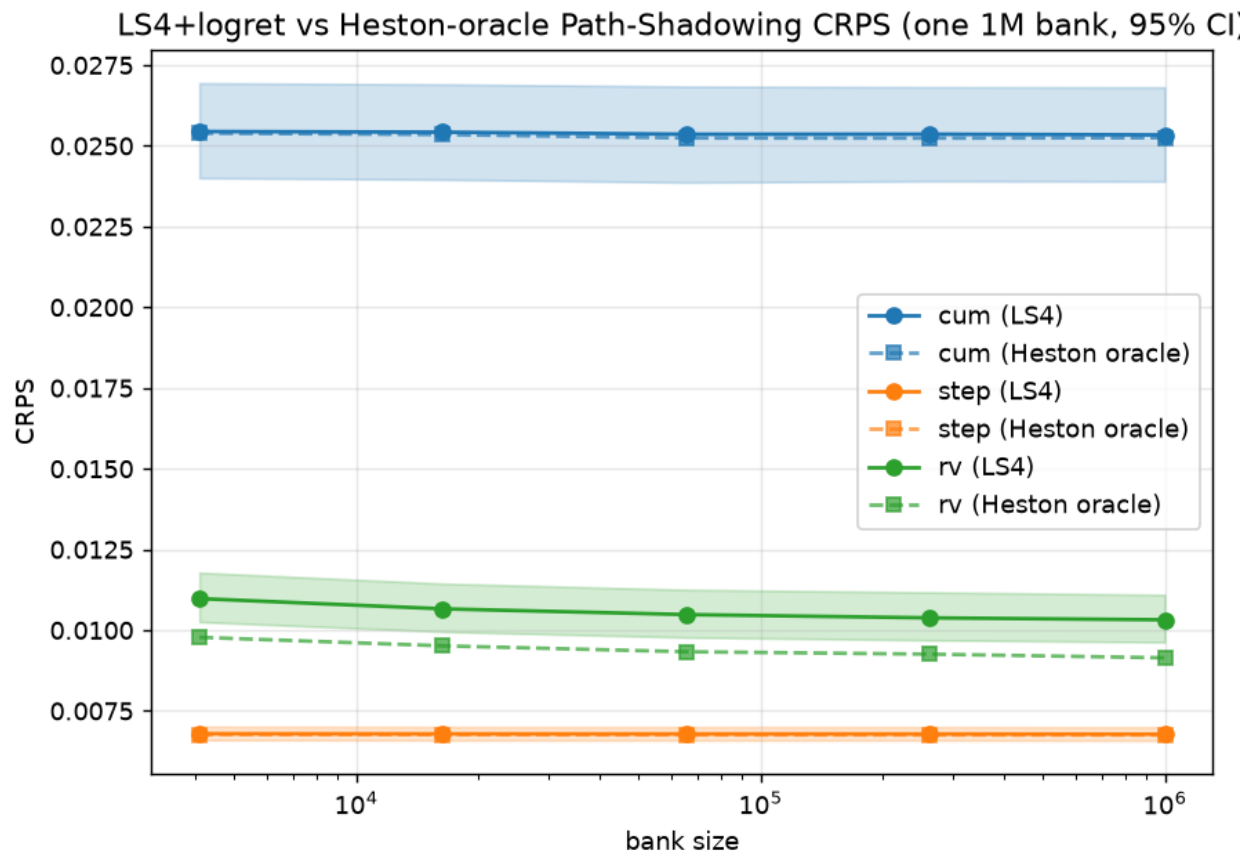


Figure 13: LS4 — CRPS vs bank size ($\log x$); solid = generator, dashed = size-matched Heston oracle, shaded = 95% bootstrap band.

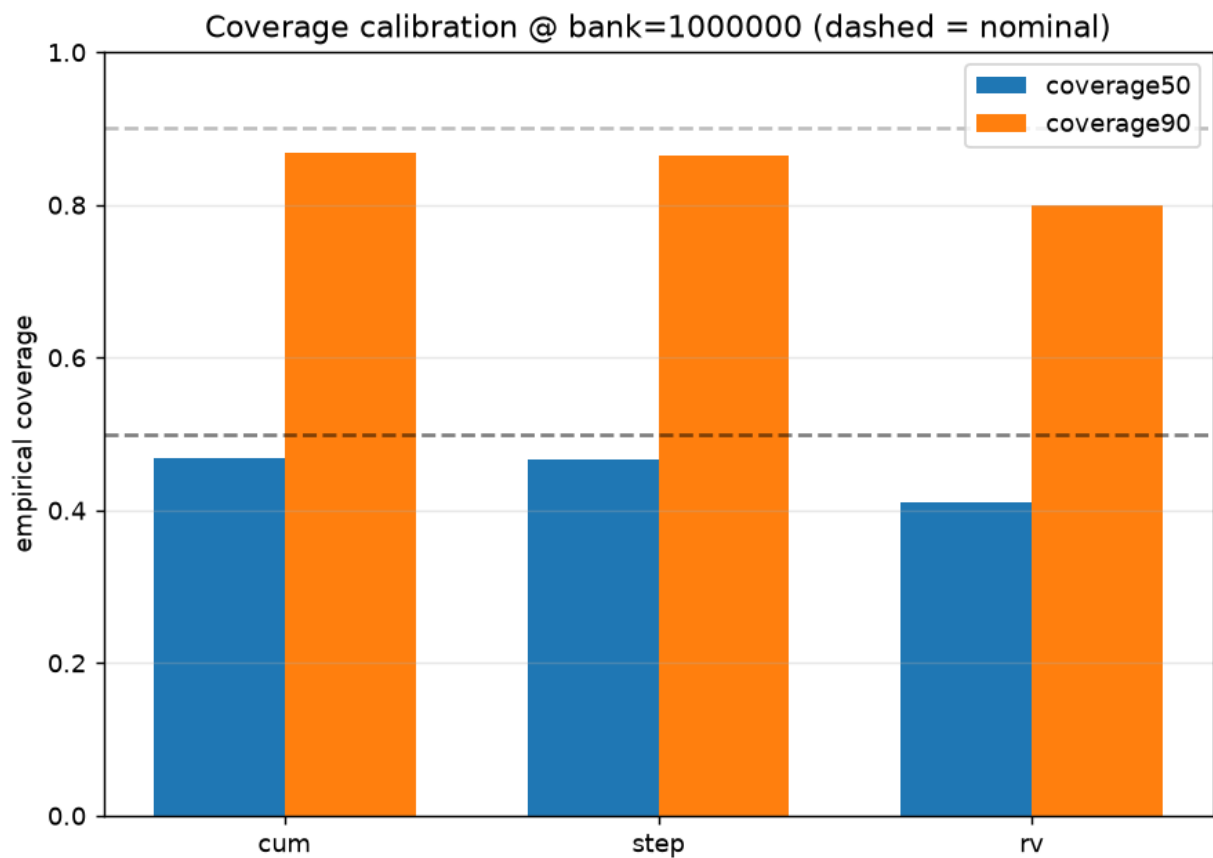


Figure 14: LS4 — Coverage calibration at the 1M bank; dashed lines = 0.50 / 0.90 nominal targets.

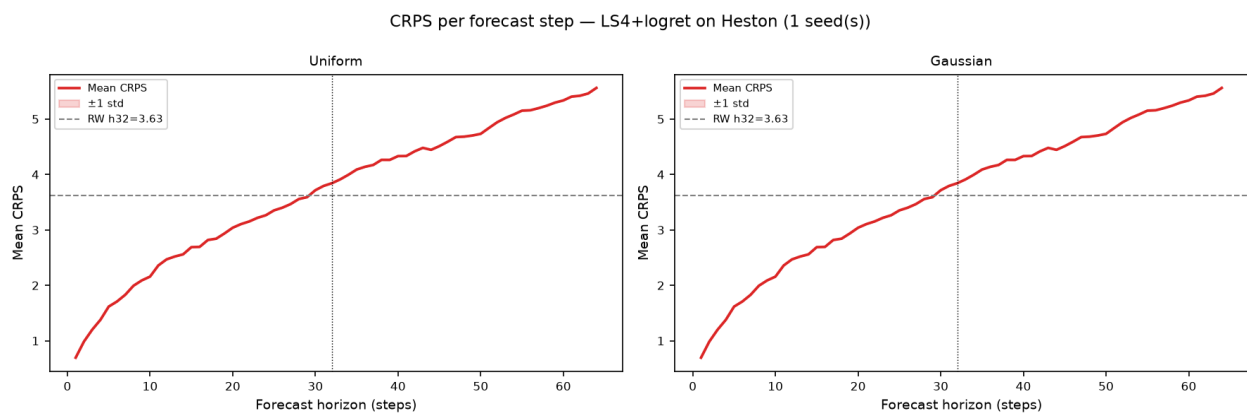


Figure 15: LS4 — CRPS per horizon step.

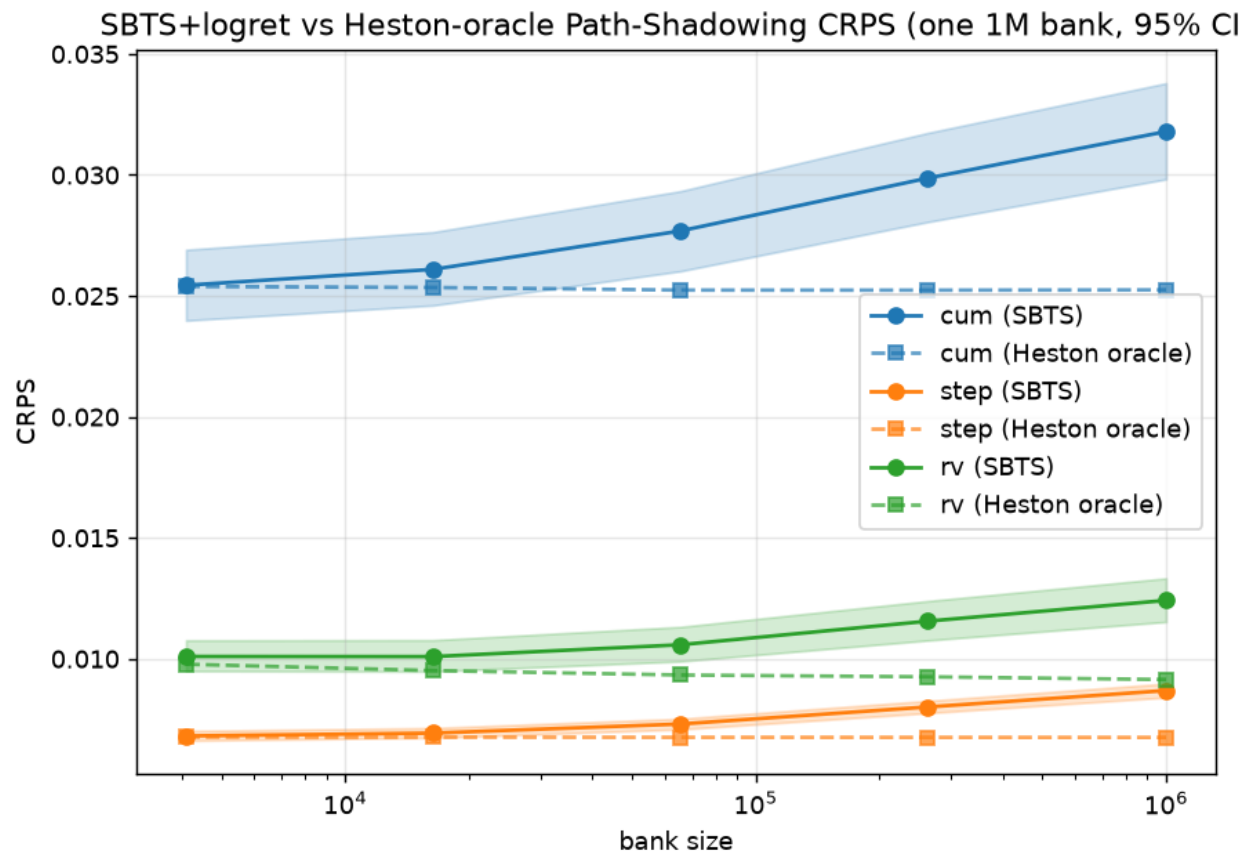


Figure 16: SBTS — CRPS vs bank size ($\log x$); solid = generator, dashed = size-matched Heston oracle, shaded = 95% bootstrap band.

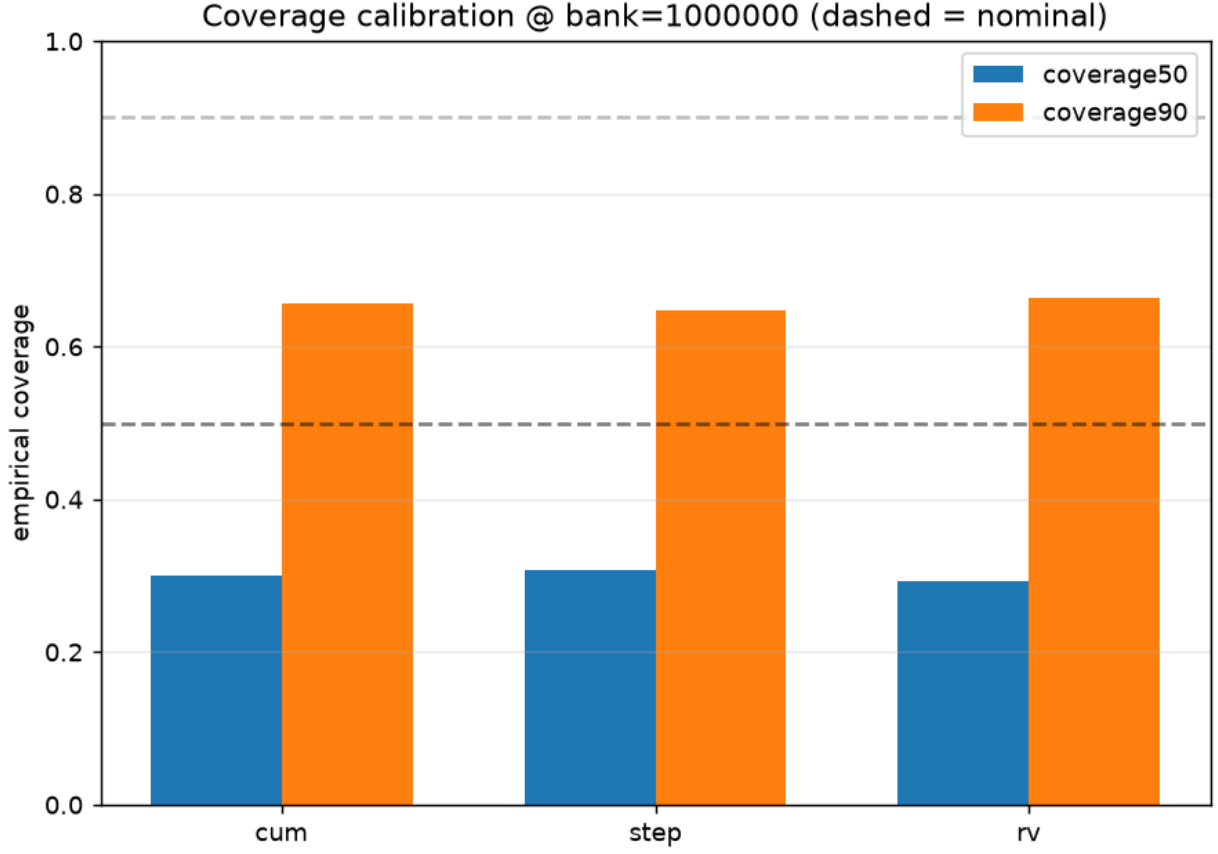


Figure 17: SBTS — Coverage calibration at the 1M bank; dashed lines = 0.50 / 0.90 nominal targets.

7.3 Training and scoring diagnostics

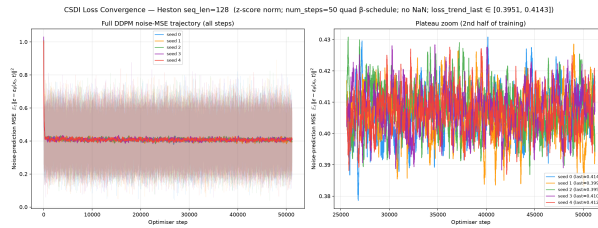


Figure 18: CSDI — training loss convergence.

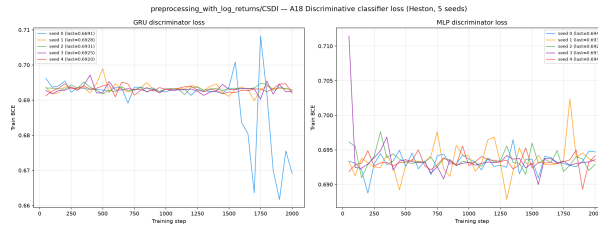


Figure 19: CSDI — discriminative-score classifier loss (A18).

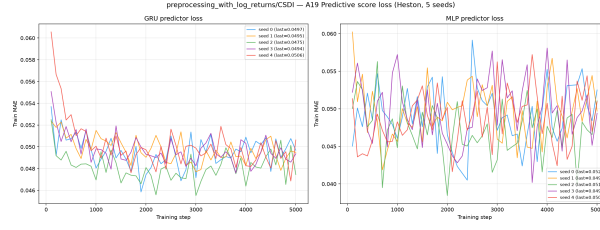


Figure 20: CSDI — predictive-score loss (A19).

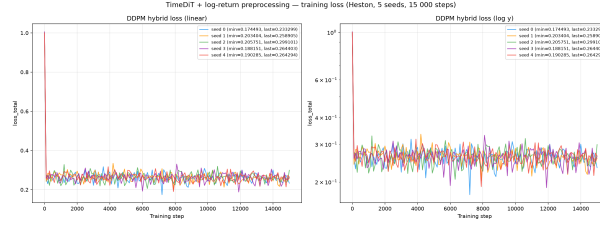


Figure 21: TimeDiT — training loss convergence.

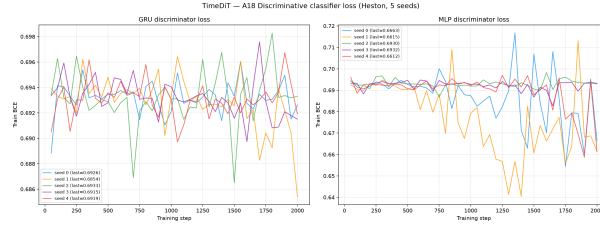


Figure 22: TimeDiT — discriminative-score classifier loss (A18).

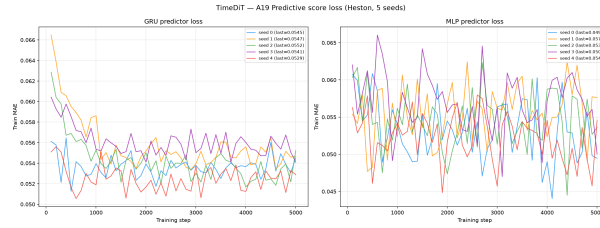


Figure 23: TimeDiT — predictive-score loss (A19).

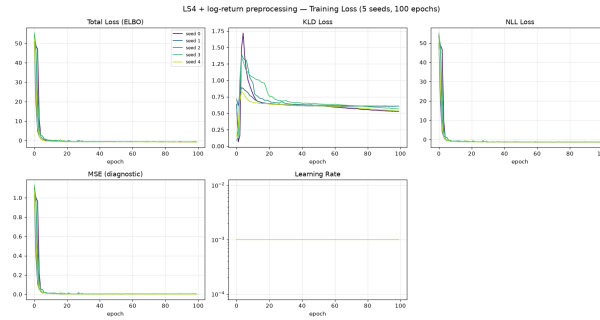


Figure 24: LS4 — training loss convergence.

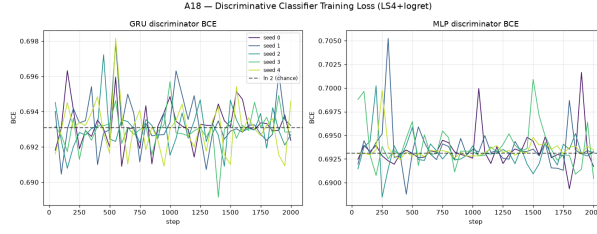


Figure 25: LS4 — discriminative-score classifier loss (A18).

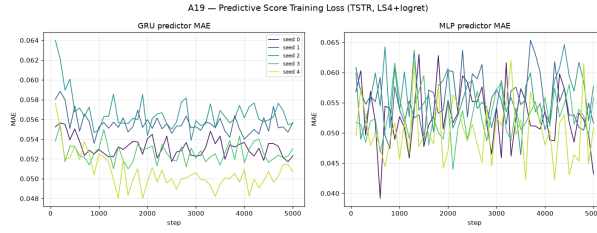


Figure 26: LS4 — predictive-score loss (A19).

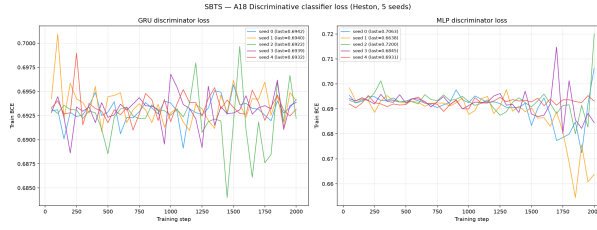


Figure 27: SBTS — discriminative-score classifier loss (A18).

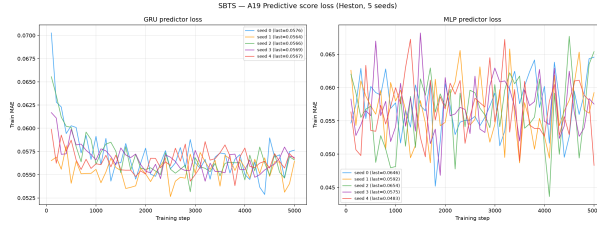


Figure 28: SBTS — predictive-score loss (A19).

7.4 Embedding visualisations (PCA / t-SNE)

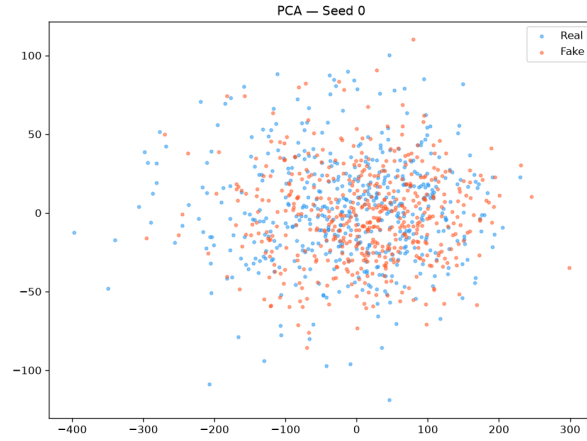


Figure 29: CSDI (log-ret) — PCA, seed 0.

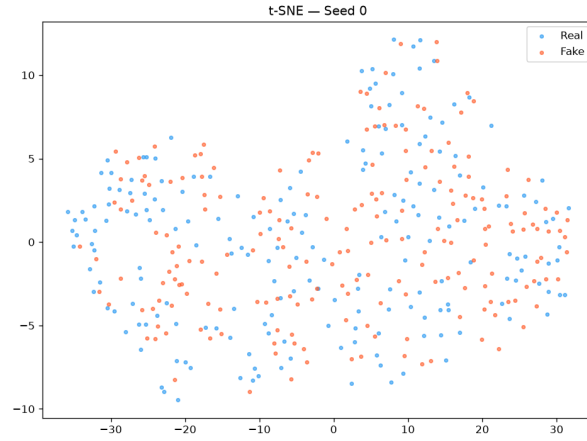


Figure 30: CSDI (log-ret) — t-SNE, seed 0.

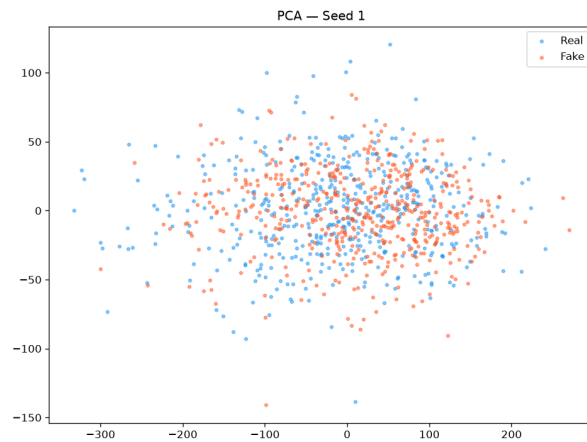


Figure 31: CSDI (log-ret) — PCA, seed 1.

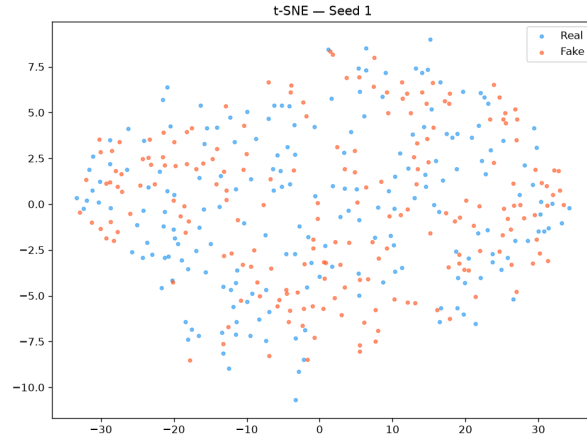


Figure 32: CSDI (log-ret) — t-SNE, seed 1.

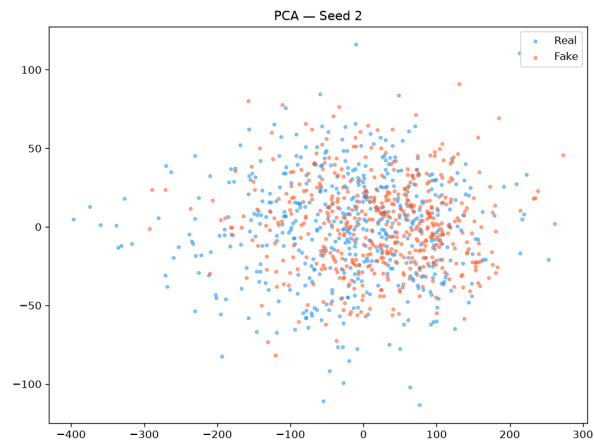


Figure 33: CSDI (log-ret) — PCA, seed 2.

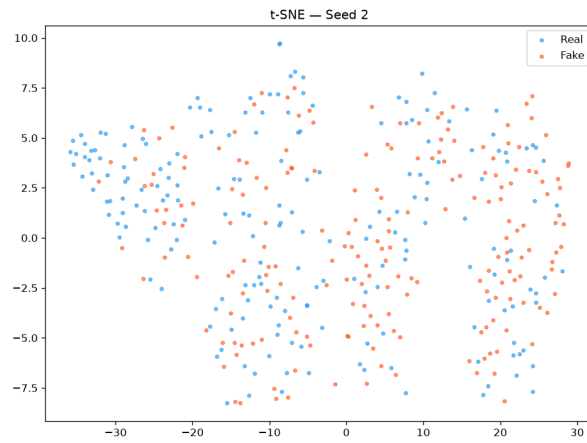


Figure 34: CSDI (log-ret) — t-SNE, seed 2.

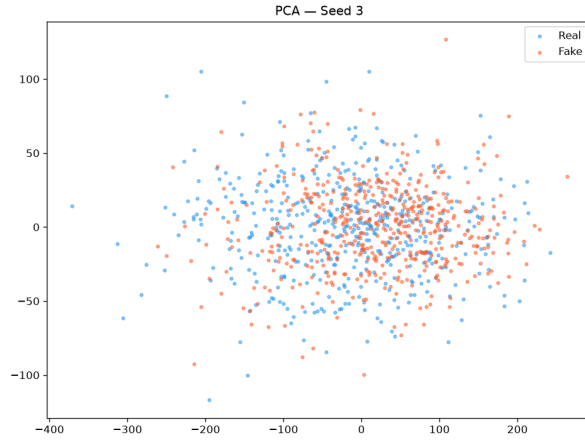


Figure 35: CSDI (log-ret) — PCA, seed 3.

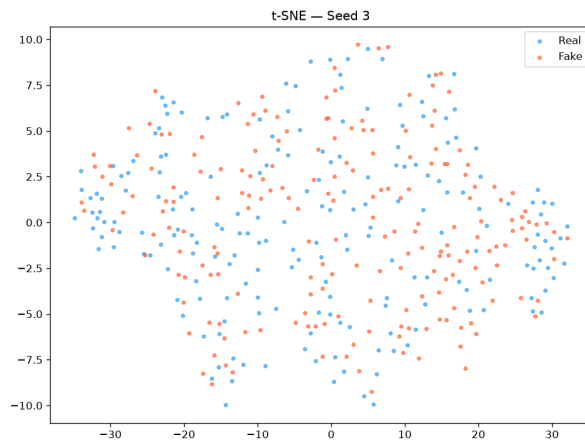


Figure 36: CSDI (log-ret) — t-SNE, seed 3.

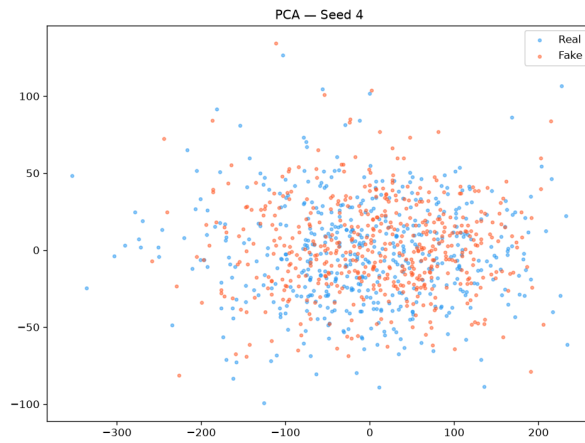


Figure 37: CSDI (log-ret) — PCA, seed 4.

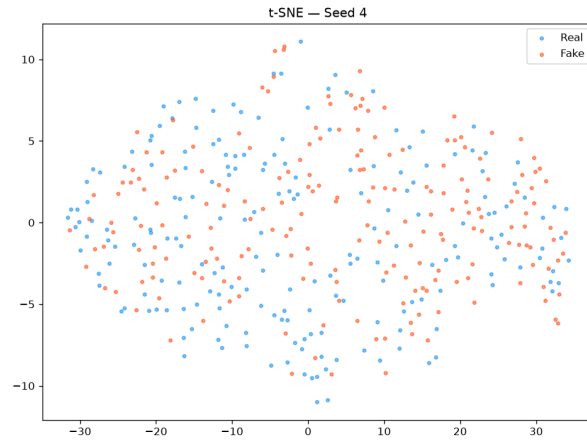


Figure 38: CSDI (log-ret) — t-SNE, seed 4.

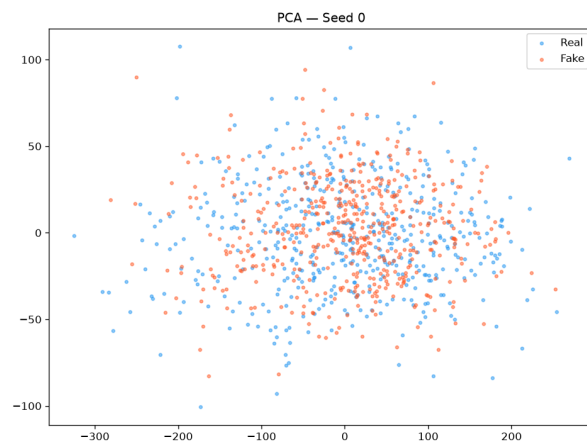


Figure 39: CSDI (raw) — PCA, seed 0.

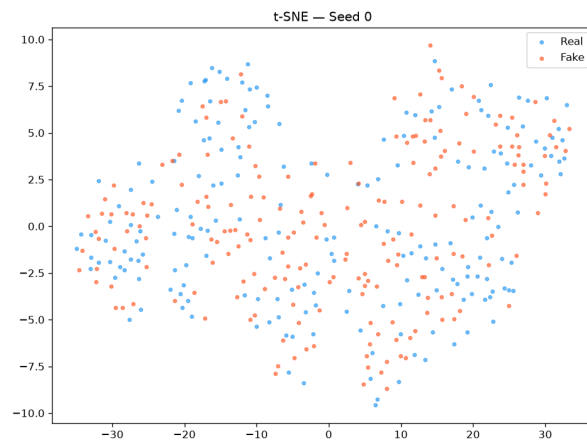


Figure 40: CSDI (raw) — t-SNE, seed 0.

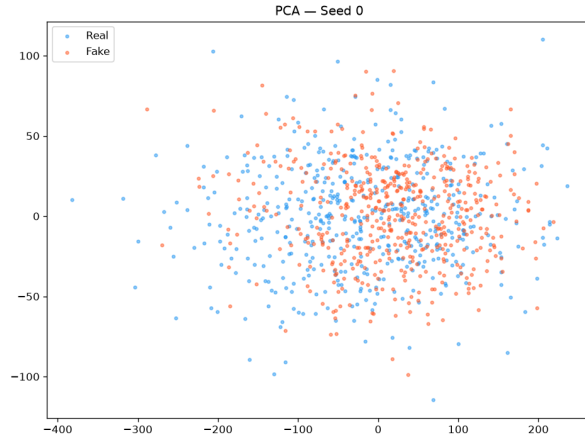


Figure 41: TimeDiT (log-ret) — PCA, seed 0.

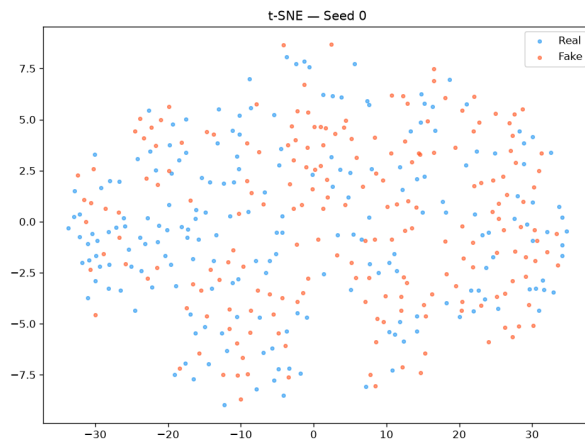


Figure 42: TimeDiT (log-ret) — t-SNE, seed 0.

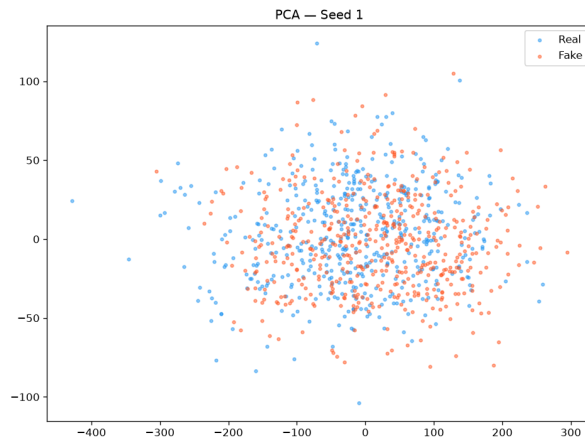


Figure 43: TimeDiT (log-ret) — PCA, seed 1.

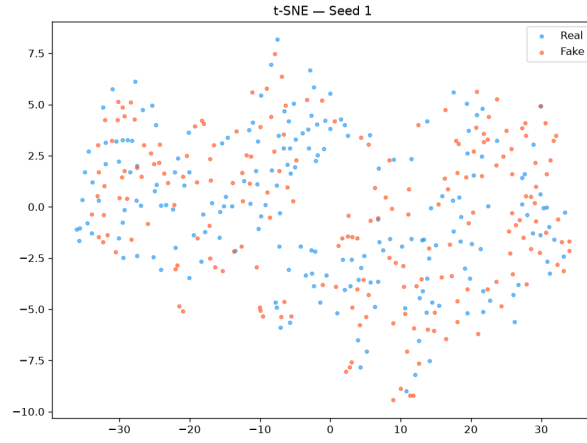


Figure 44: TimeDiT (log-ret) — t-SNE, seed 1.

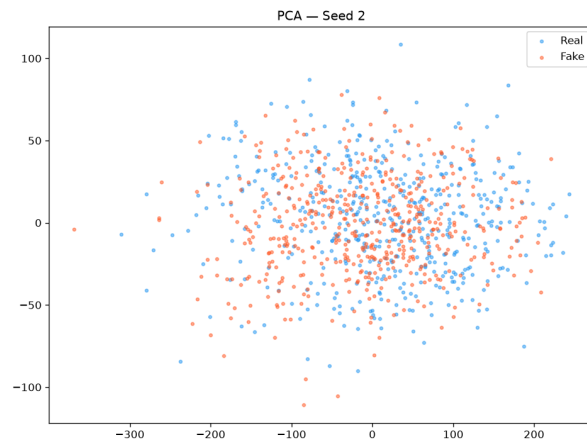


Figure 45: TimeDiT (log-ret) — PCA, seed 2.

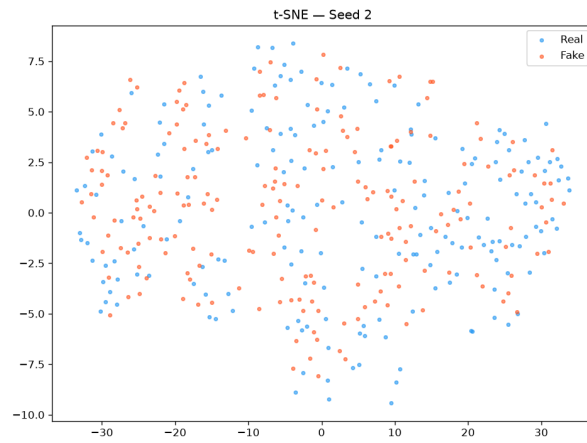


Figure 46: TimeDiT (log-ret) — t-SNE, seed 2.

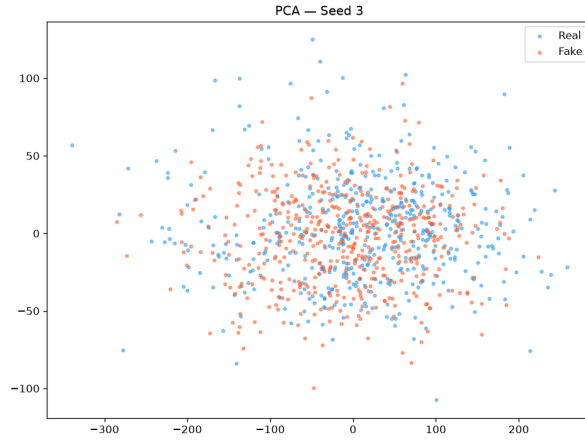


Figure 47: TimeDiT (log-ret) — PCA, seed 3.

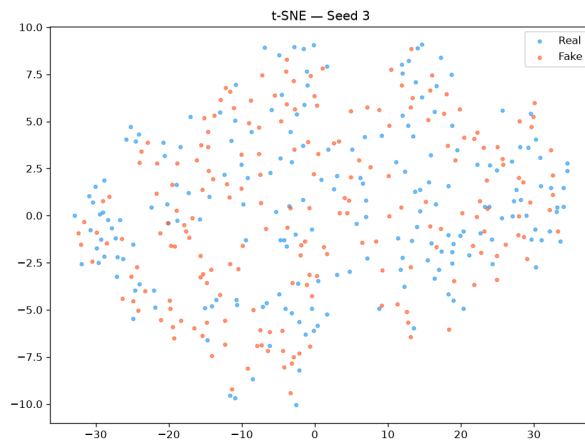


Figure 48: TimeDiT (log-ret) — t-SNE, seed 3.

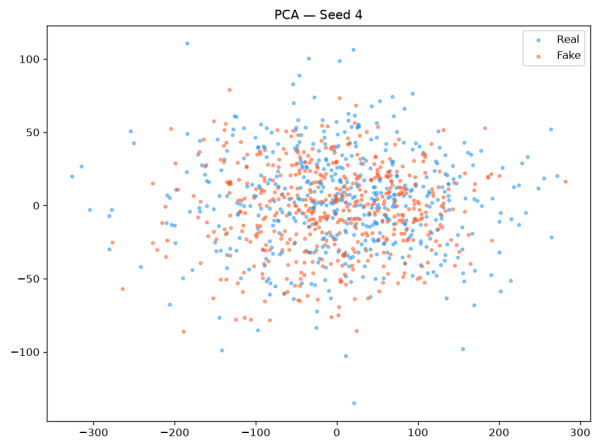


Figure 49: TimeDiT (log-ret) — PCA, seed 4.

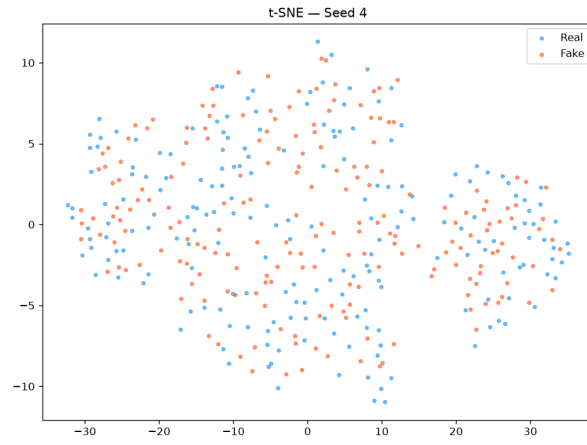


Figure 50: TimeDiT (log-ret) — t-SNE, seed 4.

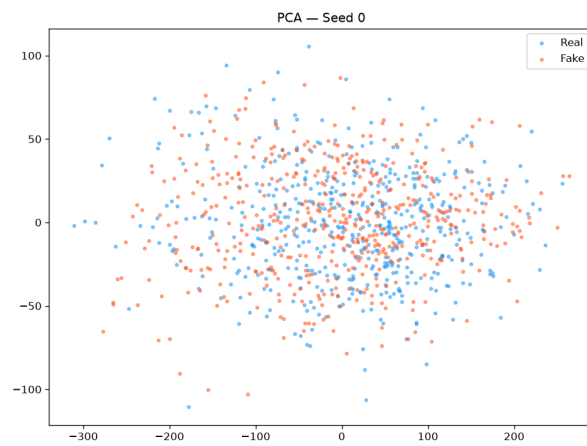


Figure 51: TimeDiT (raw) — PCA, seed 0.

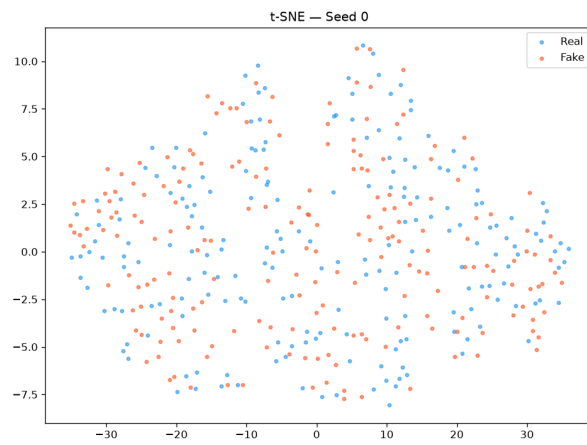


Figure 52: TimeDiT (raw) — t-SNE, seed 0.

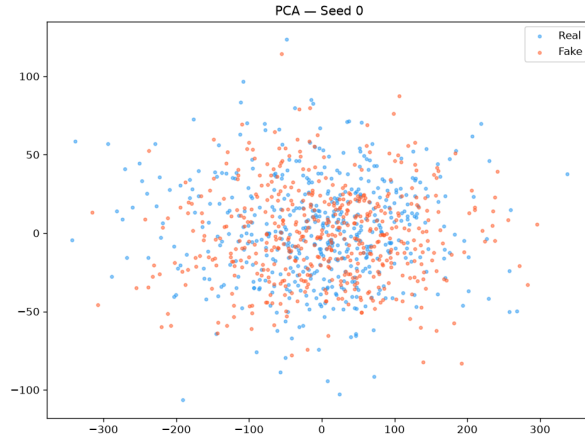


Figure 53: LS4 (log-ret) — PCA, seed 0.

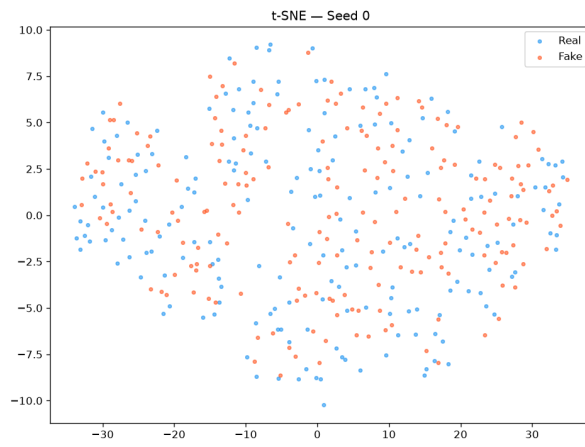


Figure 54: LS4 (log-ret) — t-SNE, seed 0.

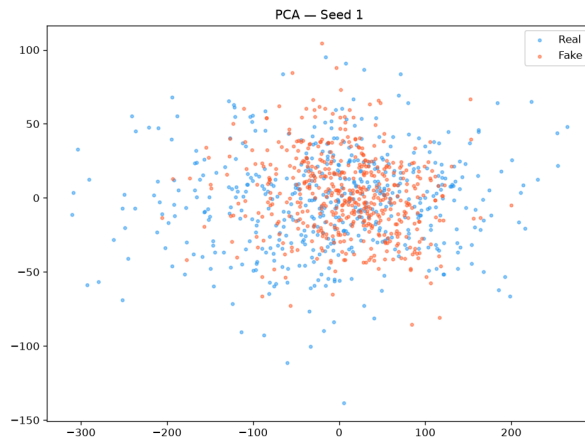


Figure 55: LS4 (log-ret) — PCA, seed 1.

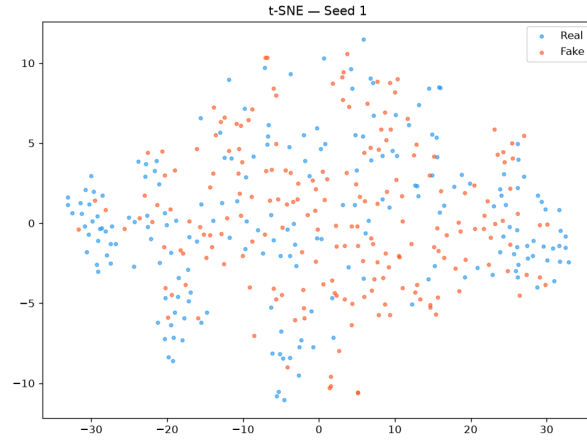


Figure 56: LS4 (log-ret) — t-SNE, seed 1.

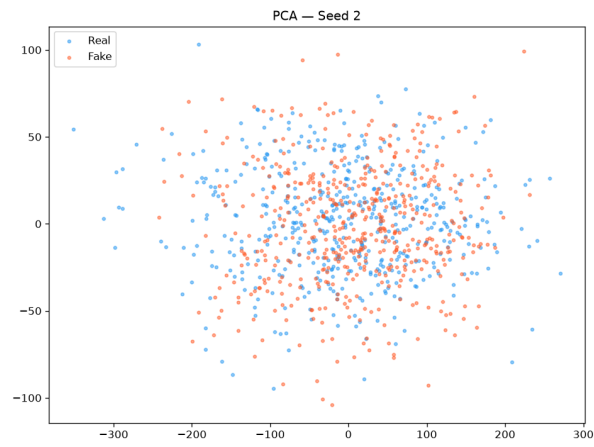


Figure 57: LS4 (log-ret) — PCA, seed 2.

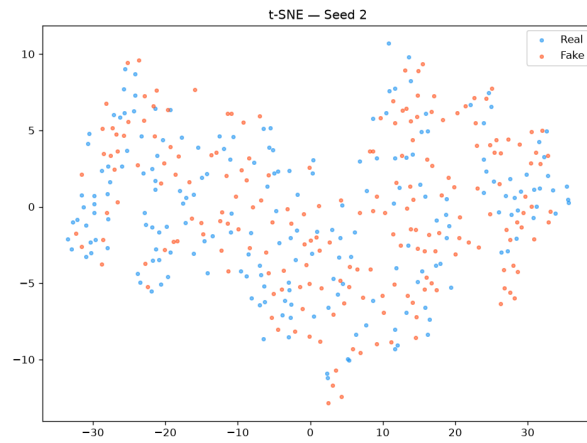


Figure 58: LS4 (log-ret) — t-SNE, seed 2.

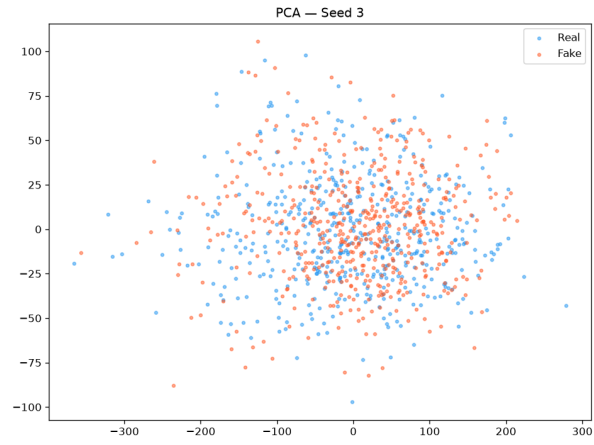


Figure 59: LS4 (log-ret) — PCA, seed 3.

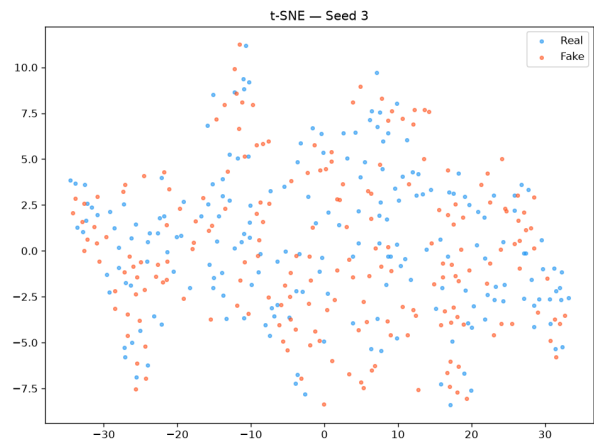


Figure 60: LS4 (log-ret) — t-SNE, seed 3.

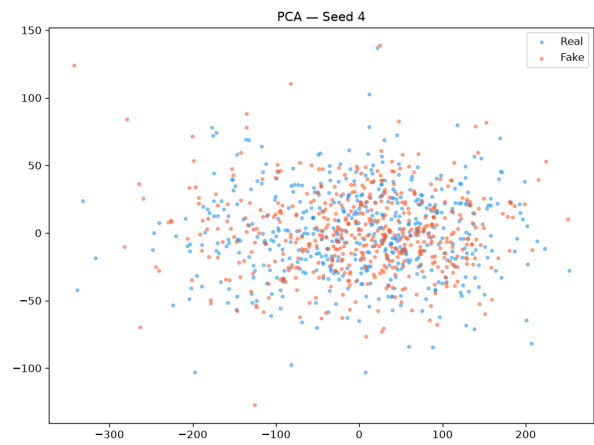


Figure 61: LS4 (log-ret) — PCA, seed 4.

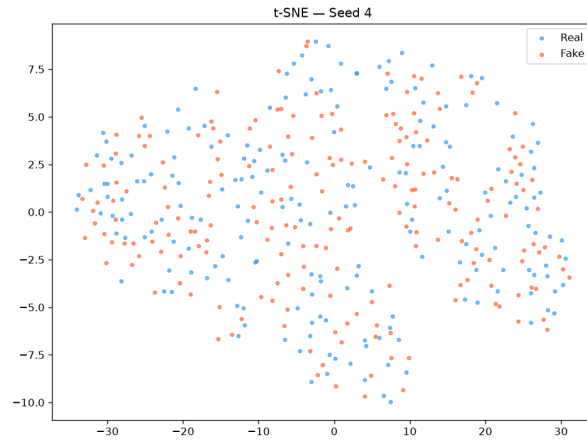


Figure 62: LS4 (log-ret) — t-SNE, seed 4.

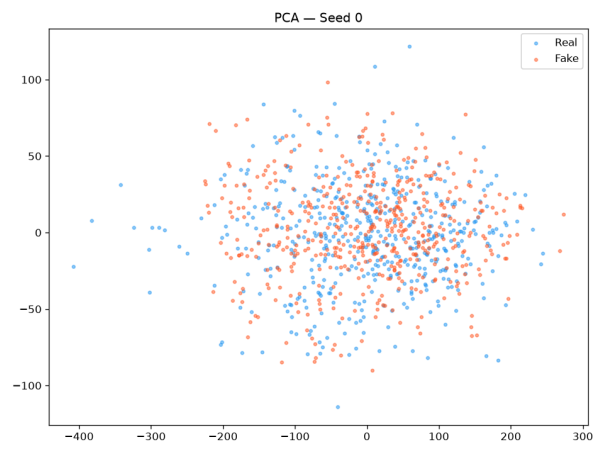


Figure 63: LS4 (raw) — PCA, seed 0.

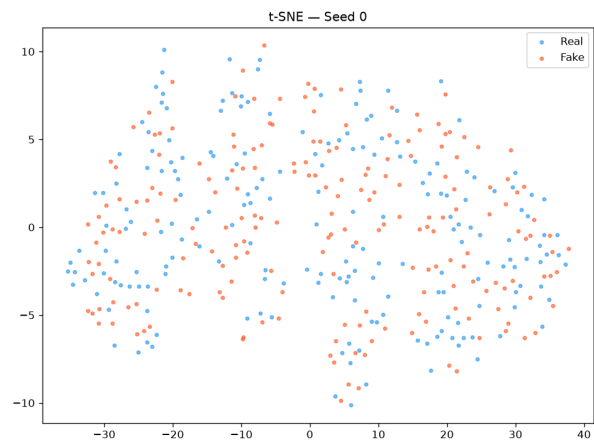


Figure 64: LS4 (raw) — t-SNE, seed 0.

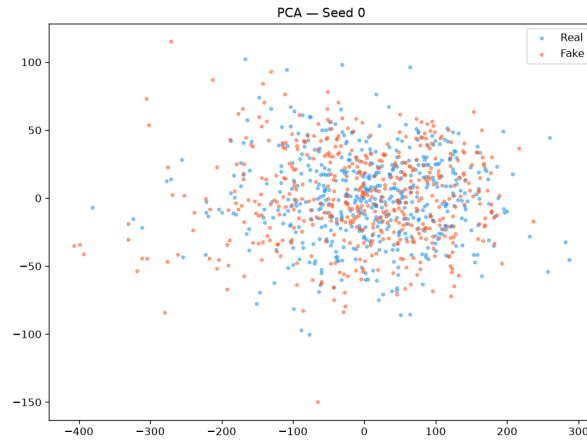


Figure 65: SBTS (log-ret) — PCA, seed 0.

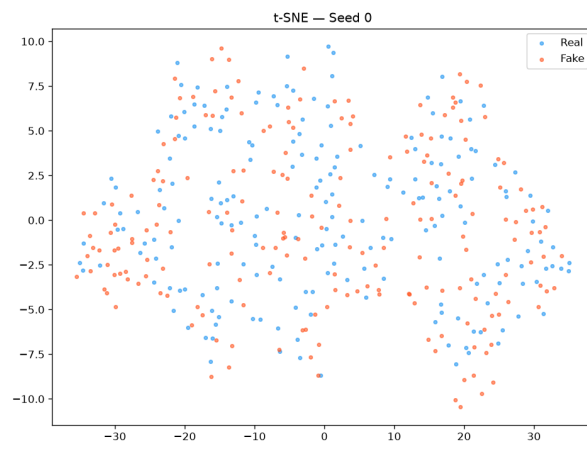


Figure 66: SBTS (log-ret) — t-SNE, seed 0.

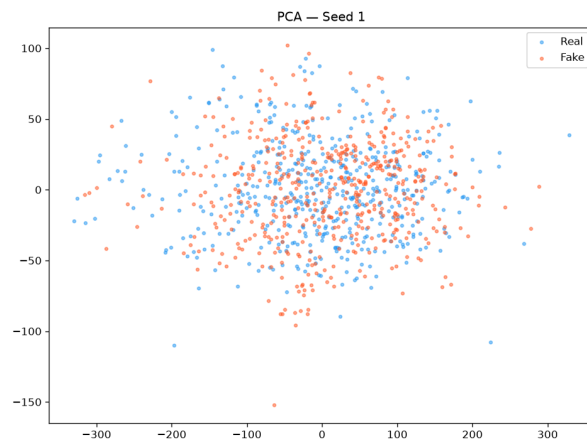


Figure 67: SBTS (log-ret) — PCA, seed 1.

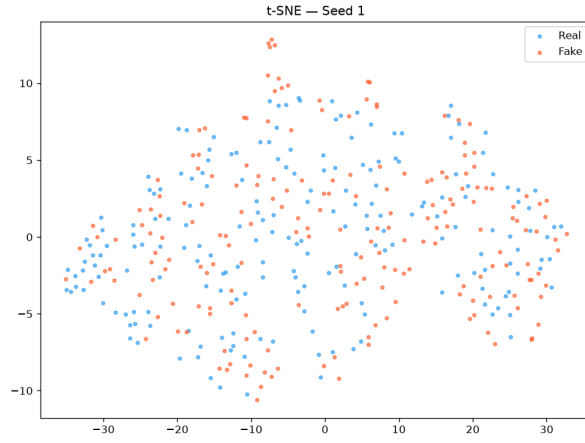


Figure 68: SBTS (log-ret) — t-SNE, seed 1.

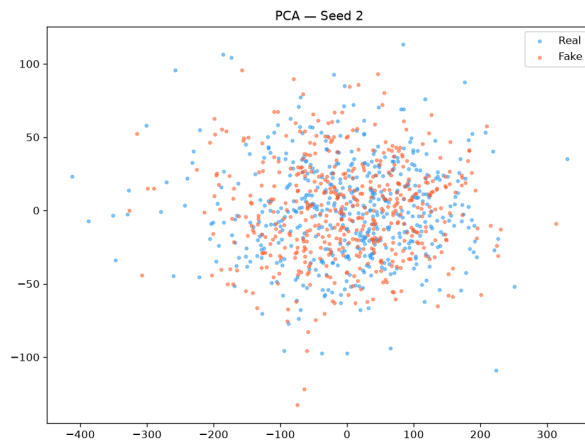


Figure 69: SBTS (log-ret) — PCA, seed 2.

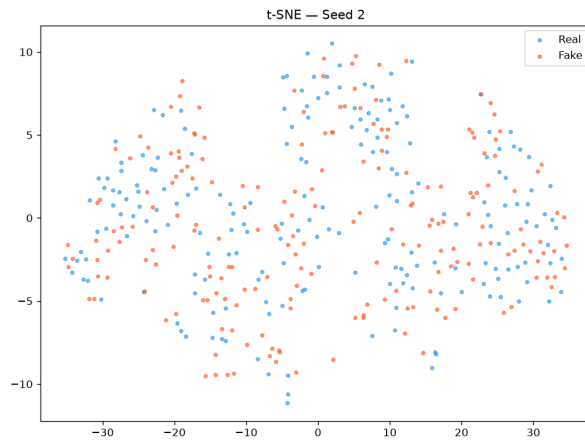


Figure 70: SBTS (log-ret) — t-SNE, seed 2.

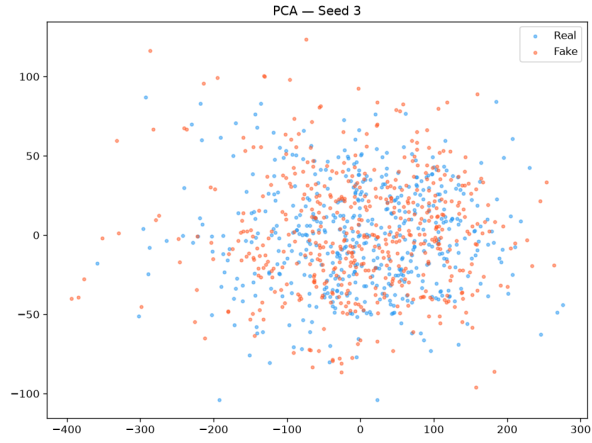


Figure 71: SBTS (log-ret) — PCA, seed 3.

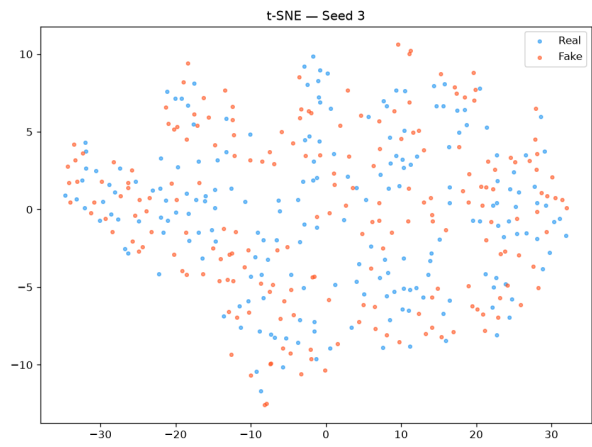


Figure 72: SBTS (log-ret) — t-SNE, seed 3.

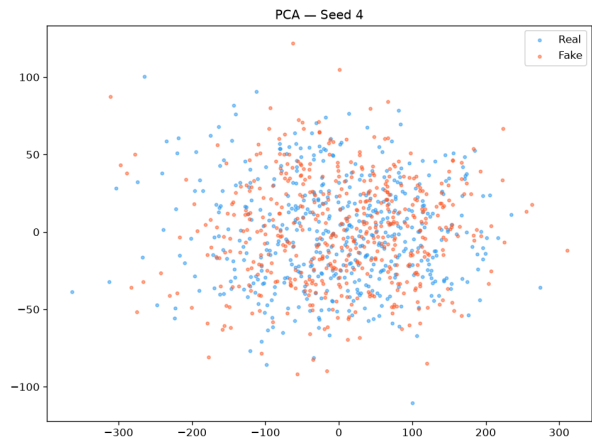


Figure 73: SBTS (log-ret) — PCA, seed 4.

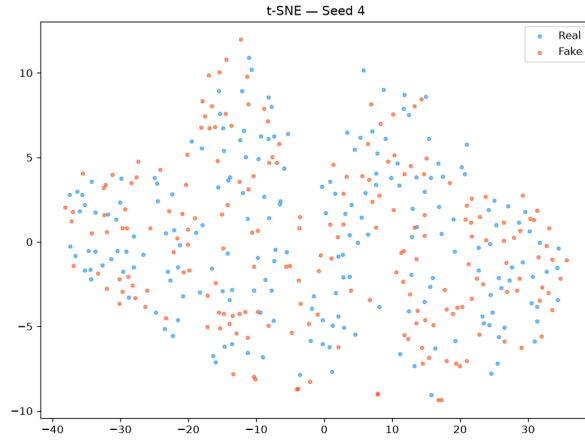


Figure 74: SBTS (log-ret) — t-SNE, seed 4.

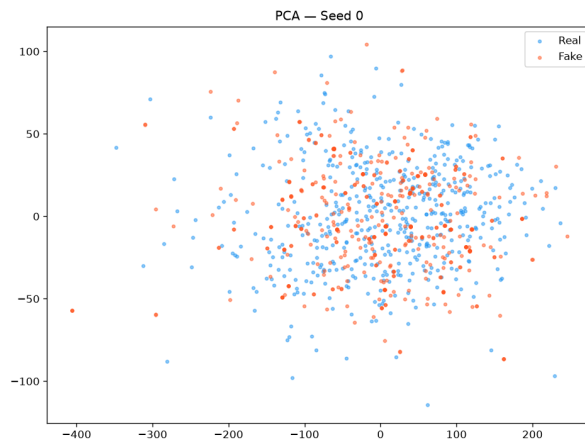


Figure 75: SBTS (raw) — PCA, seed 0.

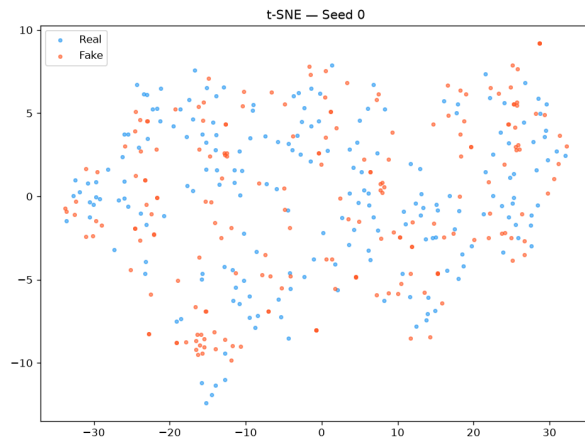


Figure 76: SBTS (raw) — t-SNE, seed 0.